4 Dynamics of Rigid Aircraft in a Nonstationary Atmosphere

4.1 Introduction

In the previous two chapters, we have described the atmospheric conditions that need to be considered for the design and operation of most aircraft, as well as the linear models that describe the low-speed response of airfoils (both supported and unsupported) to some of those conditions. We now expand our scope to study the dynamics of an aircraft flying in a nonstationary atmosphere. This is done in this chapter using the standard framework of flight dynamics, in which the aircraft is idealized as a rigid body, while subsequent chapters will consider more general situations where aeroelastic effects also need to be considered. The rigid aircraft model is still the basic foundation on which all our more general flexible aircraft models are built, and therefore it is very convenient to review first the classical theory while introducing a suitable notation that facilitates subsequent studies.

As we have discussed in Section 1.2, flight dynamics is concerned with the study of the response of the vehicle to its aerodynamic environment and (auto)pilot commands, in order to assess its performance and stability or to design control systems. It also provides the framework to describe the evolution of the aircraft with respect to an Earth-based observer, to optimize trajectories and to design maneuvers. The starting point is a suitable description of the kinematics of rigid aircraft in 3-D space, for which we define multiple coordinate systems whose relative position and orientation are sought. Spatial orientations are described in this chapter using Euler angles. Although alternative parameterizations with better numerical performance also exist, Euler angles bring more intuition into our descriptions, and we will use them extensively in this book. They are then used to derive the general (nonlinear) equations of motion (EoMs) of the rigid aircraft, which are presented in this chapter under the assumption of quasi-steady aerodynamics. That implies that the aerodynamic loading can be written as a static map (e.g., a lookup table) between the aircraft instantaneous state and the resultant forces and moments at its center of mass (CM). To consider nonstationary atmospheric conditions, aerodynamic forces are more generally written here in terms of the instantaneous relative velocity between the aircraft and a moving air mass. Stationary solutions of the EoMs determine steady-state flight conditions, which are used to define steady maneuvers and the maneuvering envelope. Finally, the linearized response of the aircraft is studied, and the response to continuous turbulence is investigated on a simplified configuration using the methods for the stochastic analysis of Section 2.3.

It is assumed throughout this book that any flight segment under consideration is of relatively short duration (typically, of the order of seconds or a few minutes), for example, a maneuver or an encounter with a thermal gust. In that case, it is appropriate to assume the "flat-Earth" approximation, thus treating the Earth surface as flat and stationary. A frame of reference on the Earth will, therefore, define an inertial reference frame in most situations in this book. Furthermore, in that frame, gravity can be assumed to be a constant vertical force that is independent of the altitude changes during the analysis. Finally, we also assume that the total mass and moment of inertia of the vehicle do not change within a given flight segment (e.g., mass of burnt fuel is negligible).

The presentation of the material in this chapter follows the structure that can be found in most classical books of flight dynamics, in particular those of Ashley (1974), Etkin (1972), and Stengel (2004). The discussion is, however, often more succinct, as our objective is to establish the basic framework that can be slowly expanded as the book progresses. In particular, most flight dynamic books include rather extensive discussions on the estimation of aerodynamic derivatives and the effect of configuration parameters on aircraft stability. Those are only superficially considered here, as we present later in this book a more general strategy in which forces on the vehicle, including unsteady and aeroelastic effects, are solved for from first principles.

4.2 Kinematics of Rigid Aircraft

4.2.1 Frames of Reference

It is necessary first to establish a convention for the description of aircraft kinematics. In this chapter, the aircraft is considered to be a rigid body, and therefore its kinematic state is uniquely determined by a measure of its position and orientation, as a function of time, from a known reference. It is then convenient to define multiple spatial reference frames whose relative positions and orientations, and velocities are sought. All reference frames in this book are defined as right-handed Cartesian bases, defined by three orthogonal unit vectors with a common origin.

Earth axes. Under the "flat-Earth" assumption, the aircraft kinematics are described with respect to an observer in an *inertial frame of reference*, that is, one that moves at most with constant translational velocity. In flight dynamics, we normally refer to this global frame of reference as the *Earth axes*.

Principal axes. The orientation of the aircraft with respect to that inertial frame of reference is given by means of a *body-attached frame of reference*, or *body axes*, that is rigidly linked to the aircraft and with origin at its CM. Most aircraft have a plane of symmetry, and it is customary to define the body-attached frame of reference such that the y axis is normal to the longitudinal symmetry plane and positive on the starboard (right) wing. In that case, y becomes a principal axis of inertia. A natural choice for the x axis is to define it as a second principal axis of inertia, as shown in Figure 4.1. Finally, the z axis completes the right-hand-rule triad, lying in the plane of

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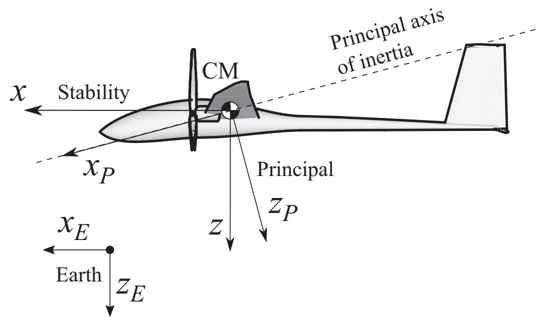

Figure 4.1 Frames of reference for a rigid aircraft: principal and stability axes (side view).

symmetry and positive downward in forward flight. The resulting body-attached frame of reference is known as the *principal axes* of the vehicle. They are uniquely defined given a vehicle geometry and mass (including payload) distribution. This frame of reference simplifies the inertia tensor, which becomes diagonal, as we will see in the following section. However, adopting this reference would unnecessarily complicate the EoMs, as this axis system would normally be tilted downward with respect to the aircraft velocity vector. This is the situation displayed in Figure 4.1.

Stability axes. A more intuitive and convenient choice for flight dynamics purposes results from aligning the x axis with the projection, on the symmetry plane, of the velocity vector of the aircraft in some nominal conditions. This defines the *stability axes*. The y–z is still the plane of symmetry, but in the reference condition, the velocity vector has no z component. In forward-flight conditions with no sideslip, which is the most common reference condition, the velocity is along the x axis. Note that the orientation of the frame with respect to the airframe changes with the point in the flight envelope (as, e.g., the aircraft may fly at different angles of incidence). Also, since the axes are body attached, the instantaneous aircraft velocity vector will not necessarily be along the x axis during maneuvers.

Wind axes. The kinematic description of both rigid and flexible aircraft dynamics requires the prior selection of a suitable frame of reference. Following the convention in flight dynamics, most derivations in this book are done using the vehicle's stability axes. However, the description of the (steady) aerodynamic characteristics of the aircraft is more easily done on a reference frame in which the x axis is always aligned with the velocity vector. This defines the *wind axes*. This frame of reference is defined such that x is always tangent to the flight path and positive forward, and z is in the plane of symmetry and positive downward as before. Wind axes are also useful when the aircraft is flying in crosswinds, as shown in Figure 4.2. Importantly, the wind axes are *not* a body-attached frame of reference.

Regarding notation, all Euclidean vectors (velocities, forces, moments, etc.) are written in this book as the column vector of their three components in a given reference frame. They are written in lowercase bold, with an uppercase subindex that indicates the frame of reference used in the projection. When necessary, we add a superscript to identify the point of application of the vector. For example, $\mathbf{v}_E^{CM}$ denotes the column

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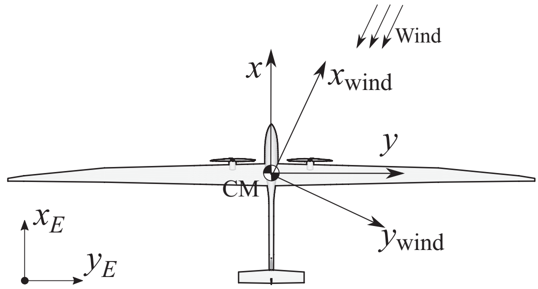

Figure 4.2 Frames of reference for a rigid aircraft: wind and stability axes (plan view).

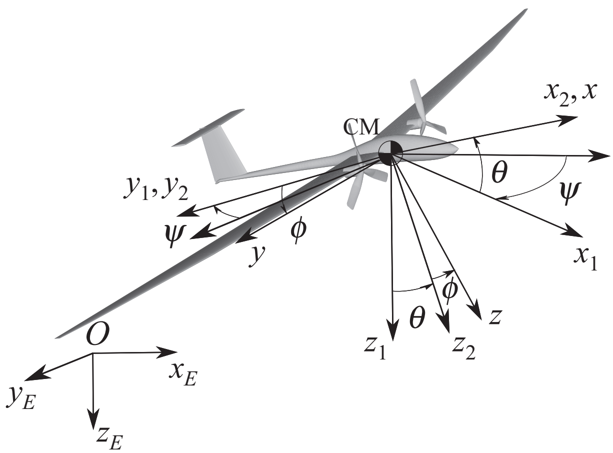

Figure 4.3 Euler angles between the Earth (origin O) and the body axes (origin at the CM).

matrix with the three components in the Earth frame of reference of the inertial velocity at the CM of the aircraft. In this chapter, most vector magnitudes are defined at the aircraft CM, however, and the CM superindex is often removed to simplify notation, such that $\mathbf{v}_E = \mathbf{v}_E^{\mathrm{CM}}$.

4.2.2 Euler Angles

Once the reference frames are selected, one needs to define their relative orientation as well as its evalution with time. Multiple alternative parameterizations (Euler or Bryant angles, Rodrigues parameters, quaternions, etc.) have been proposed for the description of finite rotations in space. A comprehensive review can be found in Bauchau (2011) and will not be repeated here. In this book, the orientation between two frames of reference and, in particular, between the Earth and the body axes is parameterized using Euler angles. As we have discussed in the introduction, this provides some insights into the mathematical formalism necessary to describe the rotation field while still using rather simple and intuitive definitions. We note, however, that quaternions have many computational advantages (as outlined, e.g., in Géradin and Cardona (2001, Ch. 4)) and are often preferred in practice.

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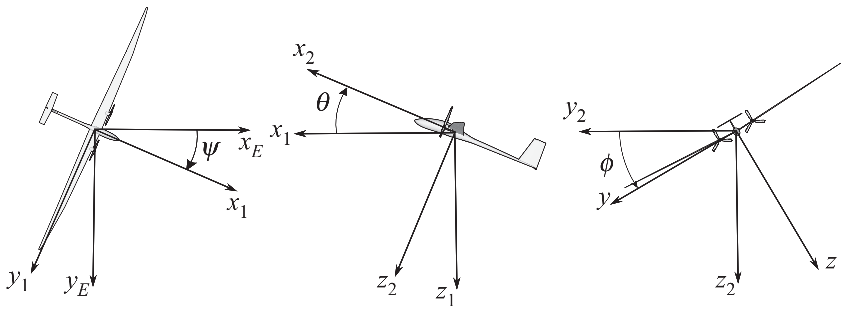

Figure 4.4 The three successive rotations defining local aircraft orientation. (a) First rotation: yaw; (b) second rotation: pitch; and (c) third rotation: roll.

Euler angles are defined by three successive rotations, starting from the Earth frame defined as in Figure 4.3, that is,

1. A rotation ψ about the vertical direction z_E of the Earth frame. ψ is the vehicle *heading* (or azimuth) and is chosen such that $-\pi \leq \psi \leq \pi$. The resulting frame of reference after the rotation is called 1 in Figure 4.3.
2. A rotation θ about the horizontal direction y_1. θ is the vehicle *pitch* (or elevation) and is chosen such that $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. The new frame of reference is called 2.
3. A rotation ϕ about the longitudinal direction x_2. ϕ, with $-\pi \leq \phi \leq \pi$, is the vehicle *bank* (or roll), and the final frame of reference defines the body-attached stability axis.

Once the order and the range of the rotations are defined, the three Euler angles that define the relative orientation between two frames of reference are unique. The convention above (yaw first, pitch second, and roll third, also known as 3-2-1), and shown in detail in Figure 4.4, is the one typically used in flight dynamics. The Euler angles allow now the transformation of coordinates of vectors in 3-D space. As an example, consider again the inertial velocity at the CM, written in terms of its components in the Earth frame, $\mathbf{v}_E$. In its components in the body-attached frame of reference, it will be $\mathbf{v}_B = \mathbf{R}_{BE}\mathbf{v}_E$, with $\mathbf{R}_{BE}$ being the *coordinate transformation matrix* from the Earth to body axes. This is obtained by the three successive rotations defined in Figure 4.4, that is,

$$\mathbf{R}_{BE} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4.1}$$

We can write it in shorthand form as $\mathbf{R}_{BE} = \boldsymbol{\tau}_x(\phi)\boldsymbol{\tau}_y(\theta)\boldsymbol{\tau}_z(\psi)$, where $\boldsymbol{\tau}_j$ identifies a rotation about the coordinate axis j. The coordinate transformation matrix from a body to the Earth can now be obtained simply by reversing the order of rotations, that is, first a rotation about x, then about y, and finally about z, with angles being equal an opposite to those defined above. Noting that each rotation is a matrix multiplication,

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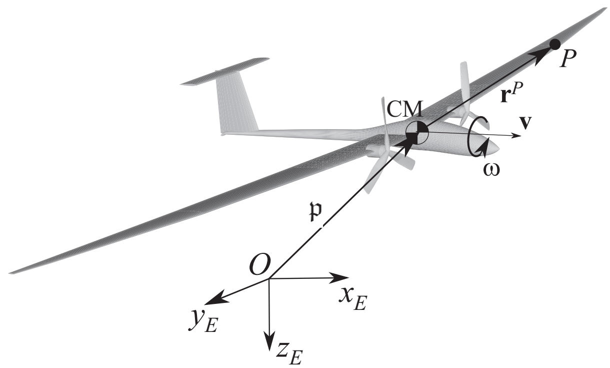

Figure 4.5 Kinematic description of a rigid aircraft.

they appear right to left in the expression as

$$\mathbf{R}_{EB} = \boldsymbol{\tau}_z(-\psi)\boldsymbol{\tau}_y(-\theta)\boldsymbol{\tau}_x(-\phi). \tag{4.2}$$

It is easy to see that $\boldsymbol{\tau}_x^\top(\phi) = \boldsymbol{\tau}_x(-\phi)$ and similarly for the other two individual rotations. As a result, the matrix defined in Equation (4.2) is the transpose of that defined in Equation (4.1), that is, $\mathbf{R}_{EB} = \mathbf{R}_{BE}^\top$. This means that by reversing the rotations, that is, applying them in the opposite order in which they were first defined, we can "undo" then. As a result, the coordinate transformation matrix is an *orthonormal matrix*, that is, its inverse is its transpose. In what follows, the inverse transformation will be simply indicated by a change of order of subindexes, as

$$\mathbf{R}_{EB} = \mathbf{R}_{BE}^{-1} = \mathbf{R}_{BE}^\top. \tag{4.3}$$

The transpose of the coordinate transformation matrix is often referred to as the *rotation matrix*. Rotations in 3-D space also play a key role in the description of the deformed shape of very flexible aircraft using geometrically nonlinear beam theory, and they will therefore appear again in Chapter 8. Further details on the properties of rotation matrices and the parameterization of rotations can be found in Appendix C.

4.2.3 Angular Velocity

Consider now the rigid aircraft of Figure 4.5. The position vector of its CM, in components in the Earth frame, is $\mathfrak{p}_E$. We are interested in describing the rotational motion of the aircraft about its CM. For that purpose, consider also an arbitrary point P in the aircraft, as shown in the figure. Its position vector with respect to the CM, expressed in its components in a certain body-attached frame of reference, B, is given by

$$\mathbf{r}_B^P = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}, \tag{4.4}$$

which, since the aircraft is considered rigid, is constant with time. When expressed in its components in the Earth reference frame, it is $\mathbf{r}_E^P(t) = \mathbf{R}_{EB}(t)\mathbf{r}_B^P$, where the

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dependence on time has been made explicit as the orientation of the aircraft changes with time. Its inertial velocity, that is, the velocity of point P with respect to an inertial observer (in particular, one in the Earth frame), can be written, in its components in the Earth reference frame, as

$$\mathbf{v}_E^P = \dot{\mathfrak{p}}_E + \dot{\mathbf{r}}_E^P = \mathbf{v}_E + \dot{\mathbf{r}}_E^P = \mathbf{v}_E + \dot{\mathbf{R}}_{EB}\mathbf{r}_B^P, \tag{4.5}$$

where $\mathbf{v}_E = \dot{\mathfrak{p}}_E$ is the inertial velocity of the CM, written in terms of its components in the Earth frame. As mentioned above, the instantaneous direction of the inertial velocity vector defines the x axis of the wind axes. An important magnitude in flight dynamics is the angle between the velocity vector and the horizontal, which is known as the *flight-path angle* (or the climb angle), γ, positive upward. Defining the unitary column vector $\mathbf{e}_3 = \left\{0 \quad 0 \quad 1\right\}^\top$, we can write

$$\gamma = -\sin^{-1}\frac{\mathbf{v}_E^\top \mathbf{e}_3}{V} \quad \text{and} \quad -\frac{\pi}{2} \le \gamma \le \frac{\pi}{2}, \tag{4.6}$$

where $V(t) = \left(\mathbf{v}_E^\top \mathbf{v}_E\right)^{1/2} = \left(\mathbf{v}_B^\top \mathbf{v}_B\right)^{1/2}$ is the magnitude of the instantaneous inertial velocity at the aircraft CM. In terms of its components in the body-attached frame of reference and recalling that $\mathbf{v}_E = \mathbf{R}_{EB}\mathbf{v}_B$, the inertial velocity of point P is given as

$$\mathbf{v}_B^P = \mathbf{v}_B + \mathbf{R}_{BE}\dot{\mathbf{R}}_{EB}\mathbf{r}_B^P. \tag{4.7}$$

From the properties given in Equation (4.3), $\mathbf{R}_{BE}\mathbf{R}_{EB} = \mathcal{I}$ is a unit matrix. This implies that $\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{R}_{BE}\mathbf{R}_{EB}) = \frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{I}) = 0$, and

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{R}_{BE}\mathbf{R}_{EB}) = \mathbf{R}_{BE}\dot{\mathbf{R}}_{EB} + \dot{\mathbf{R}}_{BE}\mathbf{R}_{EB} = \mathbf{R}_{BE}\dot{\mathbf{R}}_{EB} + \left(\mathbf{R}_{EB}^\top \dot{\mathbf{R}}_{BE}^\top\right)^\top, \tag{4.8}$$

which, using again Equation (4.3), leads to

$$\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB} + \left(\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB}\right)^\top = 0. \tag{4.9}$$

This means that $\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB}$ is a skew-symmetric matrix, and consequently that the rate of change of the rotation matrix, after being *pulled back* to B by premultiplication with $\mathbf{R}_{BE}$, is uniquely given by three independent coefficients. We then choose them as $\left(\omega_x, \omega_y, \omega_z\right)$ such that

$$\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}. \tag{4.10}$$

In this expression, we have identified the nonzero entries of $\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB}$ as the three components in body axes of a vector, which we define as the instantaneous *angular velocity* of the rigid body, $\boldsymbol{\omega}_B$. Moreover, it is very convenient to write Equation (4.10) as $\mathbf{R}_{BE}\dot{\mathbf{R}}_{EB} = \tilde{\boldsymbol{\omega}}_B$, where the operator $(\tilde{\bullet})$ is known as the *cross-product operator*, whose main properties are given in Appendix C.4. As a result, Equation (4.7) results in the well-known relation

$$\mathbf{v}_B^P = \mathbf{v}_B + \tilde{\boldsymbol{\omega}}_B\mathbf{r}_B^P. \tag{4.11}$$

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Note that in our description, we identify the components of the translational and angular velocity vectors in the body-attached frame of reference as[1]

$$\mathbf{v}_B = \begin{Bmatrix} v_x \\ v_y \\ v_z \end{Bmatrix} \quad \text{and} \quad \boldsymbol{\omega}_B = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix}. \tag{4.12}$$

Finally, we can write the angular velocity from the time derivative of the Euler angles by combining Equations (4.1) and (4.10). The algebra is rather tedious, but it is also straightforward. After going through it, the inertial angular velocity, expressed in its components in the body-attached frame of reference, can be shown as

$$\boldsymbol{\omega}_B = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} = \begin{Bmatrix} \dot{\phi} \\ 0 \\ 0 \end{Bmatrix} + \boldsymbol{\tau}_x(\phi) \begin{Bmatrix} 0 \\ \dot{\theta} \\ 0 \end{Bmatrix} + \boldsymbol{\tau}_x(\phi)\boldsymbol{\tau}_y(\theta) \begin{Bmatrix} 0 \\ 0 \\ \dot{\psi} \end{Bmatrix}. \tag{4.13}$$

Note that this result can also be obtained by the superposition of three independent angular velocities defined by the rate of change of the Euler angles around their respective axes of rotation. Those are the axes perpendicular to the planes in each of Figures 4.4a–4.4c, that is (1) an angular velocity of magnitude $\dot{\psi}$ along the Earth z axis, (2) an angular velocity of magnitude $\dot{\theta}$ along the pitching axis (axis $y_1 \equiv y_2$ in Figure 4.3), and (3) an angular velocity of magnitude $\dot{\phi}$ along the x axis of the body-attached frame of reference. This is known as the *addition theorem* (Bauchau, 2011, Ch. 4), which states that the angular velocity of a reference frame B with respect to a second frame E is the sum of the angular velocity of an arbitrary frame A with respect to E and of B with respect to A. A very important implication of the addition theorem is that angular velocities, contrary to Euler angles, are Euclidean vectors and define a vector space.

While the EoMs, could, in principle, be written using the time derivatives of the Euler angles as unknowns, the angular velocity makes for a much more compact and intuitive formulation. The transformation between both, Equation (4.13), can be written in a matrix form as

$$\boldsymbol{\omega}_B = \mathbf{T}(\phi,\theta,\psi) \begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix}, \tag{4.14}$$

where $\mathbf{T}(\phi,\theta,\psi)$ is known as the *tangential operator* corresponding to the Euler angles (see Appendix C), which is given by

$$\mathbf{T}(\phi,\theta,\psi) = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}. \tag{4.15}$$

[1] Most textbooks define the velocity components as $\mathbf{v}_B = (u, v, w)$ and $\boldsymbol{\omega}_B = (p, q, r)$. This component description can be, however, quite restrictive to extend it to flexible vehicles and it has been found convenient here to use a notation based on subindices. Note also that, unless otherwise stated, coordinates x, y, z will correspond to the instantaneous (body-attached) stability axis, as shown in Figure 4.3.

Inversion of this expression results in a set of nonlinear ordinary differential equations (ODEs) to obtain the evolution of the Euler angles given a time history of angular velocity, that is,

$$\begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} = \mathbf{T}^{-1}\boldsymbol{\omega}_B = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix}. \tag{4.16}$$

Given initial values of the Euler angles, integration in time of Equation (4.16) gives the instantaneous orientation of the rigid aircraft with respect to the Earth frame.

We have seen that any Euclidean vector, expressed in its components in an inertial reference frame, can be written in body axes by means of the coordinate transformation matrix defined in Equation (4.2). Introducing the angular velocity facilitates the computation of time derivatives. Take, for example, the velocity vector. It satisfies, $\mathbf{v}_E = \mathbf{R}_{EB}\mathbf{v}_B$, and therefore its first derivative is $\dot{\mathbf{v}}_E = \dot{\mathbf{R}}_{EB}\mathbf{v}_B + \mathbf{R}_{EB}\dot{\mathbf{v}}_B$. Premultiplying by $\mathbf{R}_{BE}$, we have finally,

$$\mathbf{R}_{BE}\dot{\mathbf{v}}_E = \dot{\mathbf{v}}_B + \tilde{\boldsymbol{\omega}}_B\mathbf{v}_B. \tag{4.17}$$

This result is known as the *transport theorem* (Greenwood, 1988). It hugely simplifies the derivation of EoMs on body-attached frames of reference, as we will see next.

4.3 Flight Dynamic Equations

4.3.1 Inertia Characteristics

Each material particle P of the vehicle, whose position vector is $\mathbf{r}_B^P$ as defined in Equation (4.4), can be associated with an infinitesimal mass dm. If $\mathcal{V}$ is the material volume of the aircraft, the total mass of the vehicle is simply

$$m = \int_{\mathcal{V}} \mathrm{d}m. \tag{4.18}$$

From the definition of the CM, we also have that $\int_{\mathcal{V}} \mathbf{r}_B^P \,\mathrm{d}m = 0$. The aircraft CM moves with some (inertial) linear and angular velocities, which in their components in body axes have been defined as $\mathbf{v}_B$ and $\boldsymbol{\omega}_B$, respectively. We define the corresponding total *linear momentum*, $\mathbf{p}_B$, as

$$\begin{aligned} \mathbf{p}_B &= \int_{\mathcal{V}} \mathbf{v}_B^P \,\mathrm{d}m = \int_{\mathcal{V}} \left(\mathbf{v}_B + \tilde{\boldsymbol{\omega}}_B\mathbf{r}_B^P\right) \mathrm{d}m \\ &= \mathbf{v}_B \int_{\mathcal{V}} \mathrm{d}m + \tilde{\boldsymbol{\omega}}_B \int_{\mathcal{V}} \mathbf{r}_B^P \,\mathrm{d}m = m\mathbf{v}_B. \end{aligned} \tag{4.19}$$

Similarly, the total *angular momentum* about the CM, $\mathbf{h}_B$, is

$$\begin{aligned} \mathbf{h}_B &= \int_{\mathcal{V}} \tilde{\mathbf{r}}_B^P \mathbf{v}_B^P \,\mathrm{d}m \\ &= \int_{\mathcal{V}} \tilde{\mathbf{r}}_B^P \left(\mathbf{v}_B + \tilde{\boldsymbol{\omega}}_B\mathbf{r}_B^P\right) \mathrm{d}m \\ &= -\tilde{\mathbf{v}}_B \int_{\mathcal{V}} \mathbf{r}_B^P \,\mathrm{d}m - \int_{\mathcal{V}} \tilde{\mathbf{r}}_B^P \tilde{\mathbf{r}}_B^P \boldsymbol{\omega}_B \,\mathrm{d}m = \mathbf{I}_B\boldsymbol{\omega}_B, \end{aligned} \tag{4.20}$$

where we have used some of the properties given in Equation (C.23) and again that $\int_{\mathcal{V}} \mathbf{r}_B^P \,\mathrm{d}m = 0$ from the definition of the CM. We have also defined here the *inertia tensor* in body axes, $\mathbf{I}_B$, which, using the cross-product operator introduced in Equation (4.10), can be written as

$$\mathbf{I}_B = -\int_{\mathcal{V}} \tilde{\mathbf{r}}_B^P \tilde{\mathbf{r}}_B^P \,\mathrm{d}m = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix}. \tag{4.21}$$

The terms in the diagonal are the moments of inertia of the body, for example, $I_{xx} = \int_{\mathcal{V}} (y^2 + z^2)\,\mathrm{d}m$. The off-diagonal terms depend on the products of inertia, for example, $I_{yz} = I_{zy} = \int_{\mathcal{V}} yz\,\mathrm{d}m$. Note that when the aircraft is symmetric about the y axis, it is $I_{xy} = 0$ and $I_{yz} = 0$.

If expressed in its components in the Earth frame, the angular momentum becomes

$$\mathbf{h}_E = \mathbf{R}_{EB}\mathbf{h}_B = \mathbf{R}_{EB}\mathbf{I}_B\mathbf{R}_{BE}\boldsymbol{\omega}_E. \tag{4.22}$$

Since by definition $\mathbf{h}_E = \mathbf{I}_E\boldsymbol{\omega}_E$, the coordinate transformation on the inertia tensor is finally given by the *congruent transformation* $\mathbf{I}_E = \mathbf{R}_{EB}\mathbf{I}_B\mathbf{R}_{BE}$. Moreover, we assume in what follows that the distribution of masses in the rigid aircraft does not change within the timescales of interest, which implies, for example, that the mass of the burnt fuel during that interval is negligible. As a result, both mass m and inertia $\mathbf{I}_B$ can be considered to be constant magnitudes with time, while $\mathbf{I}_E$ depends on the instantaneous aircraft orientation.

4.3.2 Equations of Motion of a Rigid Body

The EoM$_s$ are obtained from the Lagrange's equations, Equation (1.2), applied to a rigid body under external forces and moments. The kinetic energy is defined as $\mathcal{T} = \frac{1}{2}\int_{\mathcal{V}} \mathbf{v}_B^{P\top} \mathbf{v}_B^P \,\mathrm{d}m$, which can be written in terms of the translational and angular velocities at the CM as

$$\mathcal{T} = \frac{1}{2}m\mathbf{v}_E^\top\mathbf{v}_E + \frac{1}{2}\boldsymbol{\omega}_E^\top\mathbf{I}_E\boldsymbol{\omega}_E. \tag{4.23}$$

Substituting Equation (4.23) into Equation (1.2) results in the total applied forces equating the rate of change of linear momentum, and the applied moments about the CM equating the rate of change of angular momentum. In particular, the aircraft is subject to time-dependent external forces from which we can compute the total resultant forces and the resultant moments around the CM. Using the notation in this book, those vectors of resultant forces and moments are written in their components in the Earth frame as the column vectors $\mathbf{f}_E(t)$ and $\mathbf{m}_E(t)$, respectively. As a result, the EoMs are

$$\begin{aligned} \frac{\mathrm{d}\mathbf{p}_E}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}_E) = \mathbf{f}_E, \\ \frac{\mathrm{d}\mathbf{h}_E}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{I}_E\boldsymbol{\omega}_E) = \mathbf{m}_E. \end{aligned} \tag{4.24}$$

The first set of equations derives directly from Newton's second law, of motion while the second set is known as Euler's equations (or Euler's rotation equations). As

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a result, it is common to refer to them jointly as the *Newton–Euler equations*. As has just been mentioned, the inertia tensor in the Earth axes, $\mathbf{I}_E$, is a function of time, and it is therefore more convenient to write the equations in body axes (in particular, the stability axes). Since $\mathbf{h}_E = \mathbf{R}_{EB}\mathbf{h}_B$, we have

$$\dot{\mathbf{h}}_E = \dot{\mathbf{R}}_{EB}\mathbf{h}_B + \mathbf{R}_{EB}\dot{\mathbf{h}}_B, \tag{4.25}$$

which results again in the transport theorem, Equation (4.17). Using this relation and the transformations in Equation (4.22), we finally obtain

$$\begin{aligned} m\dot{\mathbf{v}}_B + m\tilde{\boldsymbol{\omega}}_B\mathbf{v}_B &= \mathbf{f}_{aB} + \mathbf{f}_{pB} + \mathbf{f}_{gB}, \\ \mathbf{I}_B\dot{\boldsymbol{\omega}}_B + \tilde{\omega}_B\mathbf{I}_B\boldsymbol{\omega}_B &= \mathbf{m}_{aB} + \mathbf{m}_{pB}, \end{aligned} \tag{4.26}$$

where the external forces and moments have been split among aerodynamic, propulsive, and gravitational loads. They are force and moment resultants acting on the CM and have been written in their components in the body-attached frame of reference. They are discussed next.

4.3.3 Gravitational Forces

For a given aircraft mass, m, and under the flat-Earth assumptions, the gravitational forces in body axes will solely depend on the instantaneous orientation of the vehicle. Noting that the z_E axis in the Earth frame of reference is always defined to be vertical and positive downward, the resultant gravitational force (at the CM) is

$$\mathbf{f}_{gB} = mg\mathbf{R}_{BE}\mathbf{e}_3, \tag{4.27}$$

where g is the gravitational acceleration assumed here to be constant during the time of interest. In terms of the Euler angles introduced in Equation (4.1), the components of the gravitational force in body axes are

$$\mathbf{f}_{gB} = \begin{Bmatrix} -mg\sin\theta \\ mg\cos\theta\sin\phi \\ mg\cos\theta\cos\phi \end{Bmatrix}. \tag{4.28}$$

Therefore, while the equations of motion for a rigid vehicle, Equation (4.26), could be written in terms of linear and angular velocities and their time derivatives, the gravitational forces introduce an additional explicit dependency on the vehicle orientation.[2] This means that we have to solve simultaneously Equation (4.26), which determines the six components of the linear and angular velocities, and Equation (4.16), which defines a differential equation on the Euler angles given the instantaneous angular velocity.

[2] A second potential dependency of the EoMs on the absolute spatial orientation of the vehicle occurs when the aircraft flies through a nonstationary atmosphere as is discussed in Section 4.3.5.

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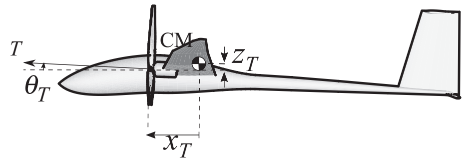

Figure 4.6 Propulsion system geometry definition.

4.3.4 Propulsive Forces

The propulsion system generates a thrust force of magnitude T on the vehicle. Thrust is typically expressed in terms of a nondimensional *thrust coefficient*, C_T, which is assumed to be a known function of the airspeed, V, and a throttle setting, δ_T (see section 2.5 of Stengel (2004) for details on typical power plants). It is defined as

$$T = \tfrac{1}{2}\rho V^2 S C_T(V, \delta_T), \tag{4.29}$$

where ρ is the local air density. In power plant selection, S is either the propeller disk area or the engine exhaust area; however, in flight dynamics modeling, it is typically the main wing reference area. For a given mass configuration of the aircraft, the instantaneous thrust will generate a resultant force and moment at the CM, which will depend on the position and orientation of the propulsive devices in the body-attached frame of reference. Assuming that the propulsion system is symmetric with respect to the aircraft x–z plane, the geometric parameters become the coordinates x_T and z_T, as well as the pitch orientation θ_T, as shown in Figure 4.6. That finally defines the propulsive forces in Equation (4.26) as

$$\mathbf{f}_{pB} = \begin{Bmatrix} T\cos\theta_T \\ 0 \\ -T\sin\theta_T \end{Bmatrix} \text{ and } \mathbf{m}_{pB} = \begin{Bmatrix} 0 \\ T(z_T\cos\theta_T + x_T\sin\theta_T) \\ 0 \end{Bmatrix}. \tag{4.30}$$

4.3.5 Quasi-Steady Aerodynamic Forces

Regarding the aerodynamic loads, specifically for this chapter, we assume that the timescales in the aircraft maneuvers are large enough such that quasi-steady aerodynamics, as defined in Section 3.3.2, can be considered. It is also assumed, as everywhere in this chapter, that the effect of any structural deformations on the aerodynamic forces can be neglected. In such a case, the aerodynamic forces on the rigid aircraft depend on the instantaneous relative velocity between the vehicle's CM and the atmosphere. We consider, in general, nonstationary atmospheric conditions, which define a varying distribution of wind velocities on the aircraft. Similar to Section 2.3.4, we, however, assume a "frozen" wind velocity field that remains unperturbed by the presence of the aircraft and that has a typical wavelength sufficiently large for the instantaneous distribution of wind velocities to vary at most linearly over the aircraft. This situation was sketched in Figure 2.17. The instantaneous wind velocity field on the moving aircraft can then be described by the time histories of two vectors of linear and angular gust velocities at the CM of the vehicle. They are naturally given with respect to the Earth frame of reference and will be referred to as $\mathbf{v}_{gE}(t)$ and $\boldsymbol{\omega}_{gE}(t)$, respectively (MIL-HDBK-1797, 1997; Richardson et al., 2013). Note that, as has already been discussed in Section 2.3.3, their particular time histories depend on both the spatial distribution of the wind field and the instantaneous orientation of the aircraft with respect to this field. The combination of both effects defines a time history of the components of both gust velocity vectors on the body-attached frame of reference (typically, the stability axes) as

$$\mathbf{v}_{gB} = \begin{Bmatrix} v_{gx}(t) \\ v_{gy}(t) \\ v_{gz}(t) \end{Bmatrix} \quad \text{and} \quad \boldsymbol{\omega}_{gB} = \begin{Bmatrix} \omega_{gx}(t) \\ \omega_{gy}(t) \\ \omega_{gz}(t) \end{Bmatrix}. \tag{4.31}$$

The aircraft CM is then selected to compute the instantaneous relative mean velocity between the vehicle and the atmosphere, which defines a nonstationary *background velocity field*. This is done for consistency with the aircraft EoMs, but any other reference point could have been equally chosen. Next, we define the aircraft instantaneous *angle of attack*, α, as the ratio of the components of the relative velocity vector in the plane of symmetry of the aircraft,

$$\alpha(t) = \tan^{-1} \frac{v_z(t) - v_{gz}(t)}{v_x(t) - v_{gx}(t)} \qquad \text{and} \qquad -\pi \leq \alpha \leq \pi. \tag{4.32}$$

Therefore, changes of the instantaneous angle of attack occur not only under changes of the CM inertial velocity but also under changes of the gust velocity. Note also that on the stability axis in calm air, the reference velocity goes along the x axis, which implies $\alpha = 0$ in forward flight using this definition.

The lateral component of the relative velocity defines the *sideslip angle*, β, that is,

$$\beta = \tan^{-1} \frac{v_y - v_{gy}}{\sqrt{\left(v_x - v_{gx}\right)^2 + \left(v_z - v_{gz}\right)^2}} \qquad \text{and} \qquad -\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2}. \tag{4.33}$$

The sideslip angle measures the lateral misalignment of the instantaneous relative velocity vector with respect to the symmetry plane of the aircraft, and it is used in the description of the aircraft lateral dynamics and when flying in crosswind. Both the angle of attack and the sideslip angle are shown in Figure 4.7.

The relative velocity at the CM between the aircraft and the atmosphere can, therefore, be alternatively defined by either the three components of the velocity vector on the (body-attached) stability axis or by its magnitude, the angle of attack, and sideslip, that is,

$$\mathbf{v}_B - \mathbf{v}_{gB} = \begin{Bmatrix} v_x - v_{gx} \\ v_y - v_{gy} \\ v_z - v_{gz} \end{Bmatrix} = V_{\text{TAS}} \begin{Bmatrix} \cos\alpha\cos\beta \\ -\sin\beta \\ \sin\alpha\cos\beta \end{Bmatrix}, \tag{4.34}$$

where $V_{\text{TAS}}(t) = \left\| \mathbf{v}_B - \mathbf{v}_{gB} \right\|$ is the true airspeed, that is, the instantaneous magnitude of the relative velocity between the aircraft and the wind. The aerodynamic forces

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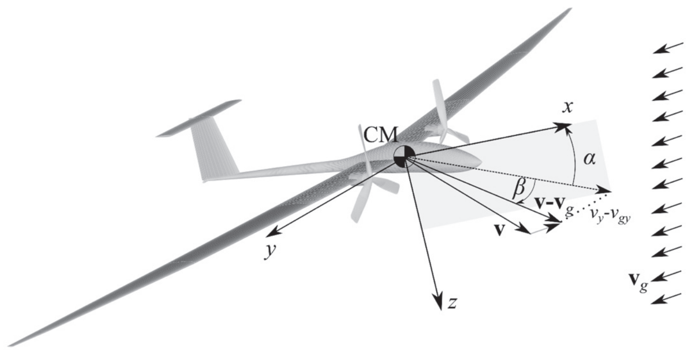

Figure 4.7 Angle of attack, α, and sideslip angle, β, for an aircraft in the presence of a constant wind velocity (not aligned with the *x* direction).

have, in general, nonzero resultants on all three components of forces and moments in Equation (4.26). The models used to obtain them vary quite substantially depending on the flight conditions and the characteristics of the vehicle. In general, we can state that they depend on three different sets of variables as follows.

First, they depend on the *vehicle geometry*, that is, magnitudes such as the wingspan, typical chord c, reference wing area S, and airfoil shape. For most aircraft, those are fixed parameters for each flight segment, while having take-off and landing configurations that differ from their cruise geometry.

Second, they depend on the *flight condition*, that is, the point of operation in the flight envelope, which determines the air density, as well as the Mach and Reynolds numbers of operation.

Third, the instantaneous aerodynamic forces depend on the relative linear and angular velocities between the vehicle and the atmosphere, and on the instantaneous deflections of any control surfaces (typically, elevators, ailerons, and rudder). If any of them varies sufficiently fast for the flow unsteadiness to become significant, then their rates (time derivatives) may need to be considered as well, as was done in Equation (3.42). However, this is not considered here, and in this chapter we use quasi-steady assumptions on the aerodynamic forces, that is, that changes in all magnitudes that define their values are sufficiently slow for the associated nondimensional frequency $\omega c/(2V)$ to be very small. This is the reduced frequency defined in Equation (3.45), which becomes again relevant here.

Consequently, for a given geometrical configuration on a vehicle, we write the aerodynamic forces and moments at the aircraft CM, expressed in their components in body axes, as

$$
\begin{aligned}
\mathbf{f}_{aB} &= \tfrac{1}{2}\rho V_{\mathrm{TAS}}^2 S \mathbf{c}_{\mathbf{f}_B}\left(\mathbf{v}_B-\mathbf{v}_{gB},\boldsymbol{\omega}_B-\boldsymbol{\omega}_{gB},\boldsymbol{\delta}_c;\mathrm{Ma},\mathrm{Re}\right),\\
\mathbf{m}_{aB} &= \tfrac{1}{2}\rho V_{\mathrm{TAS}}^2 S \boldsymbol{\Lambda}_{\mathrm{ref}} \mathbf{c}_{\mathbf{m}_B}\left(\mathbf{v}_B-\mathbf{v}_{gB},\boldsymbol{\omega}_B-\boldsymbol{\omega}_{gB},\boldsymbol{\delta}_c;\mathrm{Ma},\mathrm{Re}\right),
\end{aligned}
\tag{4.35}
$$

where $\boldsymbol{\delta}_c(t)$ is the vector with the instantaneous deflection angles in each of the control surfaces, Ma and Re are the Mach and Reynolds numbers, respectively, at the nominal conditions, $\boldsymbol{\Lambda}_{\mathrm{ref}} = \mathrm{diag}\left(b_{\mathrm{ref}}, c_{\mathrm{ref}}, b_{\mathrm{ref}}\right)$, and $\mathbf{c}_{\mathbf{f}_B}$ and $\mathbf{c}_{\mathbf{m}_B}$ are the nondimensional forces and moments after normalizing with the dynamic pressure, a reference area S, and reference lengths c_{ref} and b_{ref}. Typically, those are chosen so that S is the planform wing area, c_{ref} is the mean aerodynamic chord of the wing, and b_{ref} is the wingspan. Subindex B has been retained in both force and moment coefficients to indicate that their three components correspond to a projection on the stability axes.

Aerodynamic Derivatives

In many situations, such as at the time of stability analysis, control design, and estimation of performance, the aerodynamic forces and moments need to be known only for small changes, around a reference condition, of the control inputs, as well as the aircraft and wind velocities. The reference point is defined here as a trimmed-level flight in still air of the aircraft with velocity V, for which the reference force and moment coefficients are known and constant. Let the reference force and moment coefficients be $\mathbf{c}_{\mathbf{f}0}$ and $\mathbf{c}_{\mathbf{m}0}$, respectively. In such a case, the expressions above can be linearized about that reference, and the resulting incremental aerodynamic forces and moments are

$$\begin{aligned}
\Delta\mathbf{f}_{aB} &= \tfrac{1}{2}\rho V^2 S\left(\Delta\mathbf{c}_{\mathbf{f}_B} + 2\mathbf{c}_{\mathbf{f}0}\frac{\Delta v_x - \Delta v_{gx}}{V}\right),\\
\Delta\mathbf{m}_{aB} &= \tfrac{1}{2}\rho V^2 S\boldsymbol{\Lambda}_{\mathrm{ref}}\left(\Delta\mathbf{c}_{\mathbf{m}_B} + 2\mathbf{c}_{\mathbf{m}0}\frac{\Delta v_x - \Delta v_{gx}}{V}\right).
\end{aligned} \tag{4.36}$$

The last term in both equations comes from perturbations of the dynamic pressure about its reference value $\frac{1}{2}\rho V^2$ since V goes along the x axis in the reference condition. The incremental force and moment are also given in their components in the body-attached stability axes B in the reference condition. Next, we need to evaluate the constant matrices that determine the linear dependency between the incremental aerodynamic force and moment coefficients, $\Delta\mathbf{c}_{\mathbf{f}_B}$ and $\Delta\mathbf{c}_{\mathbf{m}_B}$, and the perturbation of $\mathbf{v}_B$, $\mathbf{v}_{gB}$, $\boldsymbol{\omega}_B$, $\boldsymbol{\omega}_{gB}$, and $\boldsymbol{\delta}_c$ from their reference values. They are known as *aerodynamic derivatives*. Noting, however, that under the quasi-steady aerodynamics assumption, forces on the aircraft only depend on its relative velocity with respect to the air mass, all dependencies with the gust velocities are equal and opposite to those of the corresponding body velocities. Therefore, only the latter ones need to be evaluated.

The convention followed in the definition of the tabulated aerodynamic derivatives comes from the classical setup in wind-tunnel testing. There, the vehicle model is positioned at an angle with respect to a constant airstream, and forces are then measured in relation to the tangent and normal directions to the flow. As a result, the quasi-steady aerodynamic characteristics of an aircraft are normally given in the wind axes, which result in the usual definitions of lift and drag as the force resultants on the negative z and x axes, respectively.[3]

[3] This is the most common convention, although not a universally accepted one. Recall that the "actual" aerodynamic forces on the aircraft are the distributed pressure and shear stress on the wet surfaces. Lift and drag are integral magnitudes that can only be measured in a wind-tunnel test through the reaction forces on the model supports.

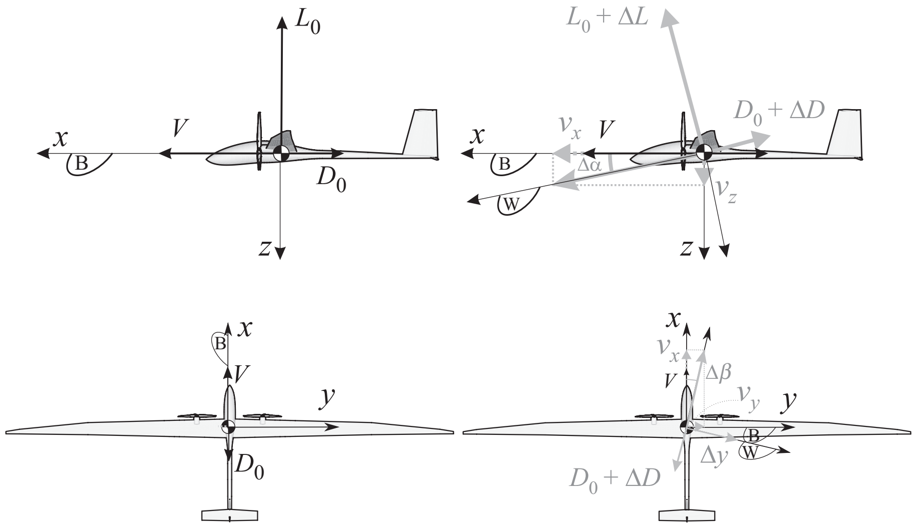

Figure 4.8 Instantaneous wind (W) and stability (B) axes in the evaluation of aerodynamic derivatives. (a) Reference condition in the $x-z$ plane, (b) perturbations in the $x-z$ plane, (c) reference condition in the $x-y$ plane, and (d) perturbations in the $x-y$ plane.

The resulting scenario is sketched in Figure 4.8. The aircraft is initially in forward flight, which defines the orientation of the frame of reference B, that is, the body-attached (stability) axes on which the EoMs are derived. A small perturbation is then considered, which results in an instantaneous velocity vector $\mathbf{v}_B(t) = \left(V + \Delta v_x(t), \Delta v_y(t), \Delta v_z(t)\right)^\top$. This defines the (incremental) angle of attack and the sideslip angle, respectively, as

$$\begin{aligned}\Delta\alpha &= \tan^{-1}\left(\frac{\Delta v_z}{V+\Delta v_x}\right) \approx \frac{\Delta v_z}{V},\\ \Delta\beta &= \tan^{-1}\left(\frac{\Delta v_y}{V+\Delta v_x}\right) \approx \frac{\Delta v_y}{V}.\end{aligned} \tag{4.37}$$

It is assumed that the angle of attack and the sideslip angle are sufficiently small such that they can, respectively, be defined in the x–z and x–y planes in the stability axes, as shown in Figure 4.8. Remember also that Δv_z is positive downward following the convention used in the definition of the stability axes (see Section 4.2.1).

The instantaneous wind axes are finally defined by means of a rotation with the (incremental) angle of attack from the reference condition, which gives the transformation matrix

$$\mathbf{T}_{WB} = \begin{bmatrix} \cos\Delta\alpha & 0 & \sin\Delta\alpha \\ 0 & 1 & 0 \\ -\sin\Delta\alpha & 0 & \cos\Delta\alpha \end{bmatrix} \begin{bmatrix} \cos\Delta\beta & \sin\Delta\beta & 0 \\ -\sin\Delta\beta & \cos\Delta\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 1 & \Delta\beta & \Delta\alpha \\ -\Delta\beta & 1 & 0 \\ -\Delta\alpha & 0 & 1 \end{bmatrix}. \tag{4.38}$$

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In that reference, aerodynamic forces and moments are obtained as

$$\begin{aligned} \mathbf{c}_{\mathbf{f}_W} &= \mathbf{T}_{WB}\mathbf{c}_{\mathbf{f}_B}, \\ \mathbf{c}_{\mathbf{m}_W} &= \mathbf{T}_{WB}\mathbf{c}_{\mathbf{m}_B}, \end{aligned} \tag{4.39}$$

where the components of both vectors have the usual definitions (Drela, 2014), that is, $\mathbf{c}_{\mathbf{f}_W} = (-C_D, C_Y, -C_L)$, the coefficients of drag, lateral force, and lift, respectively, and $\mathbf{c}_{\mathbf{m}_W} = (C_{\mathcal{L}}, C_{\mathcal{M}}, C_{\mathcal{N}})$, the coefficients of roll, pitch, and yaw aerodynamic moment, respectively. They are tabulated with respect to changes on the angle of attack, sideslip angle, flap deflections, and nondimensional angular rates. The first three have been already defined in Equations (4.32) and (4.33), and Equation (4.35), respectively, and the angular rates are also given with respect to the wind axes, that is, as three components the column vector $\boldsymbol{\omega}_W = \mathbf{T}_{WB}\boldsymbol{\omega}_B$. The most common definition is

$$\boldsymbol{\omega}_W = \frac{V}{2}\boldsymbol{\Lambda}_{\text{ref}} \begin{Bmatrix} \bar{p} \\ \bar{q} \\ \bar{r} \end{Bmatrix}, \tag{4.40}$$

where $\boldsymbol{\Lambda}_{\text{ref}}$ have been defined in Equation (4.35), and $\bar{p}$, $\bar{q}$, and $\bar{r}$ are the nondimensional roll, pitch, and yaw rates, respectively. Typical linear dependencies go as

$$\begin{aligned} C_D &\approx C_{D0} + C_{D\alpha}\Delta\alpha + C_{Dq}\Delta\bar{q} + C_{D\delta_e}\Delta\delta_e, \\ C_L &\approx C_{L0} + C_{L\alpha}\Delta\alpha + C_{Lq}\Delta\bar{q} + C_{L\delta_e}\Delta\delta_e, \\ C_{\mathcal{M}} &\approx C_{\mathcal{M}0} + C_{\mathcal{M}\alpha}\Delta\alpha + C_{\mathcal{M}q}\Delta\bar{q} + C_{\mathcal{M}\delta_e}\Delta\delta_e, \end{aligned} \tag{4.41}$$

for the longitudinal coefficients, while the lateral force and moment coefficients are written as

$$\begin{aligned} C_Y &\approx C_{Y0} + C_{Y\beta}\Delta\beta + C_{Yp}\Delta\bar{p} + C_{Yr}\Delta\bar{r} + C_{Y\delta_a}\Delta\delta_a + C_{Y\delta_r}\Delta\delta_r, \\ C_{\mathcal{L}} &\approx C_{\mathcal{L}0} + C_{\mathcal{L}\beta}\Delta\beta + C_{\mathcal{L}p}\Delta\bar{p} + C_{\mathcal{L}r}\Delta\bar{r} + C_{\mathcal{L}\delta_a}\Delta\delta_a + C_{\mathcal{L}\delta_r}\Delta\delta_r, \\ C_{\mathcal{N}} &\approx C_{\mathcal{N}0} + C_{\mathcal{N}\beta}\Delta\beta + C_{\mathcal{N}p}\Delta\bar{p} + C_{\mathcal{N}r}\Delta\bar{r} + C_{\mathcal{N}\delta_a}\Delta\delta_a + C_{\mathcal{N}\delta_r}\Delta\delta_r, \end{aligned} \tag{4.42}$$

where $C_{L\alpha} = \frac{\partial C_L}{\partial \alpha}$, $C_{Lq} = \frac{\partial C_L}{\partial \bar{q}}$, etc., with the derivatives evaluated around the reference conditions, are known as *stability derivatives*, and $C_{L\delta_e}$, $C_{Y\delta_a}$, etc. are analogously defined and known as the *control derivatives*. They are normally tabulated in terms of the Mach number and the angle of attack. It is also common to distinguish between stiffness derivatives, which are related to changes with α, β, and $\boldsymbol{\delta}_c$, and damping derivatives, associated with their rates. An excellent discussion of each of the coefficients in Equations (4.41) and (4.42) and their typical values on some conventional configurations can be found in Stevens et al. (2016, Ch. 2).

For sufficiently fast maneuvers, it is often also necessary to add to the linear expansions above the stability derivatives associated with the rates of angle of attack and control surface deflection, but those are not explicitly included here. Those are instead part of the more general formulation that we present in Chapter 7. Note that, as the angle of attack is defined as the ratio of velocities in Equation (4.32), the aerodynamic forces associated with $\Delta\dot{\alpha}$ correspond to the plunging rates defined in Section 3.3 and not to the pitch rates, in spite of the notation used. As a result, $C_{L\dot{\alpha}}$ represents the apparent mass of the fluid displaced by the acceleration of a plunging aircraft. Finally, for sufficiently fast maneuvers at transonic speeds, one needs to consider derivatives with the Mach number. Two particular cases are important. First, the changes of moment due to the shift of center of pressure that occurs in the transition between subsonic and supersonic flow, which results in the so-called *tuck derivative*, $\frac{\partial C_{\mathcal{M}}}{\partial \mathrm{Ma}}$. Second, the increase in the drag with the Mach number in that regime, $\frac{\partial C_D}{\partial \mathrm{Ma}}$, known as the *speed-damping derivative*. As the independent variable in the state vector is typically the airspeed, using Equation (4.34), the tuck and speed-damping derivatives are written in terms of V using the chain rule when added to Equation (4.41).

With this, we have that the reference force and aerodynamic coefficients in Equation (4.36) are

$$\mathbf{c_{f0}} = \begin{bmatrix} -C_{D0} \\ C_{Y0} \\ -C_{L0} \end{bmatrix} \quad \text{and} \quad \mathbf{c_{m0}} = \begin{bmatrix} C_{\mathcal{L}0} \\ C_{\mathcal{M}0} \\ C_{\mathcal{N}0} \end{bmatrix}, \tag{4.43}$$

and the incremental force and moment coefficients are finally obtained from small perturbations to Equation (4.39) as

$$\begin{aligned} \Delta \mathbf{c}_{\mathbf{f}_B} &= \frac{\partial}{\partial \Delta\alpha}\left(\mathbf{T}_{BW}\right)\mathbf{c_{f0}}\Delta\alpha + \frac{\partial}{\partial \Delta\beta}\left(\mathbf{T}_{BW}\right)\mathbf{c_{f0}}\Delta\beta + \Delta \mathbf{c}_{\mathbf{f}_B}, \\ \Delta \mathbf{c}_{\mathbf{m}_B} &= \frac{\partial}{\partial \Delta\alpha}\left(\mathbf{T}_{BW}\right)\mathbf{c_{m0}}\Delta\alpha + \frac{\partial}{\partial \Delta\beta}\left(\mathbf{T}_{BW}\right)\mathbf{c_{m0}}\Delta\beta + \Delta \mathbf{c}_{\mathbf{m}_B}, \end{aligned} \tag{4.44}$$

where $\Delta \mathbf{c}_{\mathbf{f}_B}$ and $\Delta \mathbf{c}_{\mathbf{m}_B}$ are given by the first-order approximations defined in Equations (4.41) and (4.42), respectively, and the partial derivatives of the rotation matrix with the incremental angle of attack and sideslip are given by

$$\frac{\partial}{\partial \Delta\alpha}\left(\mathbf{T}_{BW}\right) = \begin{bmatrix} -\sin\Delta\alpha & 0 & -\cos\Delta\alpha \\ 0 & 0 & 0 \\ \cos\Delta\alpha & 0 & -\sin\Delta\alpha \end{bmatrix}_{\Delta\alpha=0} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \tag{4.45}$$

$$\frac{\partial}{\partial \Delta\beta}\left(\mathbf{T}_{BW}\right) = \begin{bmatrix} -\sin\Delta\beta & -\cos\Delta\beta & 0 \\ \cos\Delta\beta & -\sin\Delta\beta & 0 \\ 0 & 0 & 0 \end{bmatrix}_{\Delta\beta=0} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{4.46}$$

Substituting Equation (4.44) back into Equation (4.36), the incremental force coefficients can be explicitly obtained in terms of the incremental linear and angular velocities of the CM and the control inputs. We will write them here as

$$\begin{Bmatrix} \Delta \mathbf{f}_{aB} \\ \Delta \mathbf{m}_{aB} \end{Bmatrix} = \rho V \boldsymbol{\mathcal{A}}_{rr} \begin{Bmatrix} \Delta \mathbf{v}_B - \Delta \mathbf{v}_{gB} \\ \Delta \boldsymbol{\omega}_B - \Delta \boldsymbol{\omega}_{gB} \end{Bmatrix} + \frac{1}{2}\rho V^2 \boldsymbol{\mathcal{A}}_{ru} \boldsymbol{\delta}_c, \tag{4.47}$$

where $\boldsymbol{\mathcal{A}}_{rr}$ and $\boldsymbol{\mathcal{A}}_{ru}$ are the constant matrices of *rigid-body aerodynamic influence coefficients* associated with the states and inputs in the problem. They depend on the geometry, the stability and control derivatives, and the flight regime (reference Reynolds and Mach numbers). The estimation of stability and control derivatives for rigid aircraft, Equations (4.41) and (4.42), is extensively described in most textbooks on flight dynamics (e.g., Stengel, 2004, Ch. 3) and will not be discussed here. Computationally efficient methods are also now available to construct them from CFD simulations at the vehicle level, such as the one proposed by Da Ronch et al. (2013) using harmonic balance methods. Section 4.6 discusses an example of the derivation of the aerodynamic influence coefficients in Equation (4.47) under simplified conditions, while more general linear aerodynamic models, valid also for flexible aircraft, will be described in Section 6.5.1.

4.4 State-Space Description

We have so far derived the EoMs that describe the flight dynamics of a rigid aircraft and the main features of the external forces on the vehicle. From a mathematical point of view, the aircraft can be seen as a dynamical system; we are concerned about its equilibrium conditions (trimmed flight, static maneuvers), stability (its ability to return to the equilibrium when it is perturbed), and maneuverability (response to commanded inputs). Those problems are presented in the final sections of this chapter. To facilitate those studies, we first "repackage" in this section the EoMs into a state-space formulation.

4.4.1 State and Input Vectors

First, we define the *state vector* for the rigid aircraft dynamics, $\mathbf{x}_r(t)$, as

$$\mathbf{x}_r = \begin{Bmatrix} v_x & v_y & v_z & \omega_x & \omega_y & \omega_z & \phi & \theta \end{Bmatrix}^\top. \tag{4.48}$$

The state vector includes all the necessary information to track future vehicle kinematics given its current values and the future input time histories. We have not included the heading, ψ, as under the flat-Earth assumption, this value has no effect in the vehicle dynamics – we say that the dynamics is *invariant* with respect to ψ.

Next, we proceed to identify the possible actions on the system. From Section 4.3.4, we have seen that the throttle setting, δ_T, is the command associated with the power plant. In general, we may have multiple, independently actuated engines, but for simplicity, we use a single input signal in our discussion. Equally, the aerodynamic forces given by Equation (4.35) include commanded inputs by means of the control surface deflections, $\boldsymbol{\delta}_c$. We assume a single command for elevators, δ_e, ailerons. With this, we can define the *input vector*, $\mathbf{u}(t)$, as

$$\mathbf{u} = \begin{Bmatrix} \delta_e & \delta_a & \delta_r & \delta_T \end{Bmatrix}^\top. \tag{4.49}$$

Finally, the nonstationary atmospheric conditions, as seen by the aircraft at any given instant, define the *disturbance vector*, $\mathbf{w}(t)$, which is defined for convenience as

$$\mathbf{w} = \begin{Bmatrix} -v_{gx} & -v_{gy} & -v_{gz} & -\omega_{gx} & -\omega_{gy} & -\omega_{gz} \end{Bmatrix}^\top. \tag{4.50}$$

With this, we can write the dynamics of the system, given by Equations (4.16) and (4.26), in their most generic form as

$$\begin{aligned} \dot{\mathbf{x}}_r &= \mathfrak{f}(\mathbf{x}_r, \mathbf{u}, \mathbf{w}) \\ &= \mathfrak{f}_{\text{gyr}}(\mathbf{x}_r) + \mathfrak{f}_{\text{aero}}(\mathbf{x}_r, \mathbf{u}, \mathbf{w}) + \mathfrak{f}_{\text{prop}}(\mathbf{x}_r, \mathbf{u}, \mathbf{w}) + \mathfrak{f}_{\text{grav}}(\mathbf{x}_r). \end{aligned} \tag{4.51}$$

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Figure 4.9 Basic block diagram for the aircraft state equations.

The term $\mathfrak{f}_{\text{gyr}}$ includes the gyroscopic forces in the problem, which are the terms with the cross products of the angular velocity in Equation (4.26). For convenience, we also include there the right-hand side of the kinematic equations, Equation (4.16). In Equation (4.51), we have also made explicit that the gyroscopic terms only depend on the state of the system. The aerodynamic forcing terms are $\mathfrak{f}_{\text{aero}} = \left\{ \mathbf{f}_{aB}^\top \quad \mathbf{m}_{aB}^\top \quad 0 \quad 0 \right\}^\top$, with the forces and moments that have been defined in Equation (4.35). Note that this brings a dependency of the equations on the flight condition (altitude and flight Mach number). Similar expressions define the propulsion and gravitational terms, that is, $\mathfrak{f}_{\text{prop}}$ and $\mathfrak{f}_{\text{grav}}$, from Equations (4.28) and (4.30), respectively. Equation (4.51) is normally complemented by an output equation that extracts magnitudes of interest into a generic *output vector*, $\mathbf{y}$ (for instance, to interface with a flight simulator). In its most general form, the resulting system is written as

$$\begin{aligned} \dot{\mathbf{x}}_r &= \mathfrak{f}(\mathbf{x}_r, \mathbf{u}, \mathbf{w}), \\ \mathbf{y} &= \mathfrak{g}(\mathbf{x}_r, \mathbf{u}, \mathbf{w}). \end{aligned} \tag{4.52}$$

The output function, $\mathfrak{g}$, is often defined as a linear combination of the states, and often the values of certain states themselves, but it could also include magnitudes defined by nonlinear relations, such as the instantaneous airspeed, $V = \sqrt{v_x^2 + v_y^2 + v_z^2}$.

Equation (4.52) needs to be solved together with the initial condition, that is, a set of inputs, $\mathbf{u}(0) = \mathbf{u}_0$, states, $\mathbf{x}_r(0) = \mathbf{x}_{r0}$, and wind conditions, $\mathbf{w}(0) = \mathbf{w}_0$, at time $t = 0$. The flow diagram describing Equation (4.52) is the simple input–output relation, as shown in Figure 4.9.

4.4.2 Small-Perturbation Equations

We have seen that the EoMs for rigid aircraft can be written as a set of nonlinear state-space equations. Most analyses use as a reference an equilibrium point of the aircraft (*trim conditions*), defined by combinations of inputs $\mathbf{u}_0$, states $\mathbf{x}_{r0}$, and constant wind conditions $\mathbf{w}_0$ such that

$$\mathfrak{f}(\mathbf{x}_{r0}, \mathbf{u}_0, \mathbf{w}_0) = \mathbf{0}. \tag{4.53}$$

This equilibrium could be in straight-level flight or in a steady maneuver, which are discussed, for the particular case $\mathbf{w}_0 = \mathbf{0}$, in Section 4.5. Small-perturbation analysis on the dynamic equations around this reference condition is useful to analyze the stability characteristics of this equilibrium point, as well as to define (linear) feedback control strategies. For that purpose, we approximate the time histories around an equilibrium condition as

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$$\begin{aligned}\mathbf{x}_r(t) &\approx \mathbf{x}_{r0} + \Delta\mathbf{x}_r(t),\\ \mathbf{u}(t) &\approx \mathbf{u}_0 + \Delta\mathbf{u}(t),\\ \mathbf{w}(t) &\approx \mathbf{w}_0 + \Delta\mathbf{w}(t),\end{aligned} \tag{4.54}$$

where $\Delta\mathbf{x}_r(t)$, $\Delta\mathbf{u}(t)$, and $\Delta\mathbf{w}(t)$ are assumed to be sufficiently small. Substituting Equation (4.54) into the first line in Equation (4.52), and subtracting the reference condition, given in Equation (4.53), result in

$$\Delta\dot{\mathbf{x}}_r = \mathfrak{f}\left(\mathbf{x}_{r0} + \Delta\mathbf{x}_r, \mathbf{u}_0 + \Delta\mathbf{u}, \mathbf{w}_0 + \Delta\mathbf{w}\right) - \mathfrak{f}\left(\mathbf{x}_{r0}, \mathbf{u}_0, \mathbf{w}_0\right). \tag{4.55}$$

The right-hand side of this equation gives the incremental forcing terms corresponding to the small perturbations defined above. In particular, if we invoke the form of Equation (4.51), it can be written as $\Delta\mathfrak{f} = \Delta\mathfrak{f}_{\text{gyr}} + \Delta\mathfrak{f}_{\text{aero}} + \Delta\mathfrak{f}_{\text{prop}} + \Delta\mathfrak{f}_{\text{grav}}$, although this is not explicitly used now, and it is left instead for the next sections. Using its first-order Taylor expansion, Equation (4.55) can be approximated as

$$\begin{aligned}\Delta\dot{\mathbf{x}}_r &= \left.\frac{\partial\mathfrak{f}}{\partial\mathbf{x}_r}\right|_0 \Delta\mathbf{x}_r + \left.\frac{\partial\mathfrak{f}}{\partial\mathbf{u}}\right|_0 \Delta\mathbf{u} + \left.\frac{\partial\mathfrak{f}}{\partial\mathbf{w}}\right|_0 \Delta\mathbf{w}\\ &= \mathbf{A}_{rr}\Delta\mathbf{x}_r + \mathbf{B}_r\Delta\mathbf{u} + \mathbf{E}_r\Delta\mathbf{w},\end{aligned} \tag{4.56}$$

which is a continuous-time LTI state-space representation of the rigid aircraft dynamics, with the system dynamics given as in Section 1.4.2 but now also with disturbance inputs coming from the wind velocity. In Equation (4.56), we have defined the rigid aircraft state matrix $\mathbf{A}_{rr}$, the input matrix $\mathbf{B}_r$, and the disturbance input matrix $\mathbf{E}_r$ at the trim condition. Detailed expressions for each of them are obtained in Section 4.6 around a steady climb. Contributions of the aerodynamic forces and moments are of particular importance. From Equation (4.47), it can be seen that the three terms in the linearized aerodynamics define contributions to the state, input, and disturbance matrices in Equation (4.56).

This linearization is only valid locally around a reference condition, and therefore the system matrices are typically parameterized by four independent variables that define steady-state flight conditions for a given aircraft configuration (Stengel, 2004), namely altitude (which determines air density), true airspeed (the velocity of the aircraft relative to the mean wind), which is often given in terms of the Mach number, bank angle (in rolling maneuvers), and climb angle (in climbing maneuvers).

4.5 Steady-State Flight

We have by now built the basic framework to investigate the aircraft flight dynamics, which can now be explored. We are first concerned with the *maneuverability* of rigid-wing aircraft, that is, their ability to follow a desired flight path, and then with the *maneuver loads*, that is, the internal stresses that appear in the airframe due to the external forces necessary to alter the vehicle trajectory. In general, we distinguish between *steady* and *dynamic maneuvers* depending on whether the aircraft linear and angular velocity components in the body-attached reference frame are constant with time or are time dependent. We restrict ourselves here to steady maneuvers that are further assumed to be performed in calm air. Steady maneuvers, or *steady-steady flight*, are used to define the nominal maneuvering envelope of a given vehicle (Section 4.5.2), while also determining the reference conditions for dynamic stability analysis and the initial conditions for dynamic events, such as the response of the aircraft to nonstationary atmospheric conditions. Those are studied in subsequent sections in this chapter.

4.5.1 Steady Maneuvers

Steady maneuvers correspond to solutions of Equation (4.26) with constant translational and angular velocity in the body-attached reference frame, that is,

$$\begin{aligned} \mathbf{f}_{aB} + \mathbf{f}_{pB} + \mathbf{f}_{gB} - m\tilde{\boldsymbol{\omega}}_B \mathbf{v}_B &= 0, \\ \mathbf{m}_{aB} + \mathbf{m}_{pB} - \tilde{\boldsymbol{\omega}}_B \mathbf{I}_B \boldsymbol{\omega}_B &= 0, \end{aligned} \tag{4.57}$$

which are solved together with the kinematic relations defined by Equation (4.16). The first set of equations here represents the equilibrium of forces along each of the three body-fixed axes. The second set, likewise, represents the moment equilibrium about each of the body-fixed axes. All six equations must be satisfied simultaneously for the aircraft to be considered in equilibrium (or trim). The problem to be solved is then finding the control inputs, for which Equations (4.57) are identically satisfied. This results in five different steady maneuvers that can be achieved by an air vehicle, namely straight-level flight ($\dot{\phi} = \dot{\theta} = \dot{\psi} = 0, \phi = \theta = 0$), steady climb ($\dot{\phi} = \dot{\theta} = \dot{\psi} = 0, \phi = 0$), steady pull-up ($\dot{\phi} = \dot{\psi} = 0, \phi = 0$), steady turn ($\dot{\phi} = \dot{\theta} = 0$), and steady roll ($\dot{\theta} = \dot{\psi} = 0$). The first four, shown schematically in Figure 4.10, are considered in vehicle design and are discussed next. Note that straight-level flight is a particular case of a steady climb.

Steady Climb

In a *steady climb*, the aircraft follows a rectilinear trajectory at constant speed, that is, the nonzero terms in the state vector are $v_x = V$ and $\theta = \theta_0$, which are both constant with time. Note also that in this case, $\theta = \theta_0 = \gamma$, which is the flight-path angle (see Equation (4.6)). Considering only the set of force equilibrium equations now, we can use that the external forces in the body-fixed frame based on the stability axes are

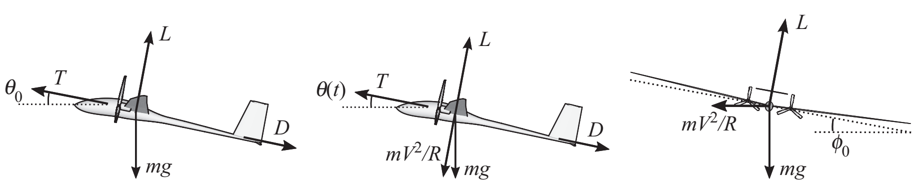

Figure 4.10 Steady Maneuvers. (a) Steady climb, (b) steady pull-up, and (c) steady turn.

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$$\mathbf{f}_{aB} = \begin{Bmatrix} -D \\ 0 \\ -L \end{Bmatrix}, \quad \mathbf{f}_{pB} = \begin{Bmatrix} T\cos\theta_T \\ 0 \\ -T\sin\theta_T \end{Bmatrix}, \quad \text{and} \quad \mathbf{f}_{gB} = \begin{Bmatrix} -mg\sin\theta_0 \\ 0 \\ mg\cos\theta_0 \end{Bmatrix}. \tag{4.58}$$

The first force equilibrium equation from the set in Equation (4.57) results in

$$D - T\cos\theta_T + mg\sin\theta_0 = 0, \tag{4.59}$$

where we used the fact that $\omega_B = 0$. The second force equilibrium equation is trivially satisfied, while the third force equation is given by

$$L + T\sin\theta_T - mg\cos\theta_0 = 0. \tag{4.60}$$

The characteristics of the climb are closely linked to the available thrust. The key metric used for this is the *specific excess power*, which measures the difference between the maximum available power and the power necessary to maintain level flight at a given airspeed and altitude (Stengel, 2004).

Steady Pull-Ups and Turns

In steady pull-up and turn maneuvers, the aircraft follows a circular trajectory in either a vertical (pull-up) or a horizontal (turn) plane. Let R be the radius of the trajectory in either case. The corresponding centripetal forces have been included in the free-body diagrams of Figure 4.10b and 4.10c. In the case of *steady pull-up*, the nonzero states are $v_x = V$ and $\omega_y = \frac{V}{R}$, both constant. Since $\omega_y = \dot{\theta}$, the pitch angle varies with time as $\theta(t) = \theta_0 + \frac{V}{R}t$. The nontrivial force equilibrium equations in this case along the x and z directions, respectively, are

$$\begin{aligned} L + T\sin\theta_T - mg\cos\theta(t) - m\frac{V^2}{R} &= 0, \\ T\cos\theta_T - D - mg\sin\theta(t) &= 0. \end{aligned} \tag{4.61}$$

This clearly requires that lift (and therefore drag) and thrust be constantly updated to sustain the maneuver. Finally, the *steady turn* is similarly defined with a constant angular rate $\frac{V}{R}$ in the horizontal plane and a constant roll angle ϕ_0. As the stability axes are body attached, this results in an angular velocity with two nonzero components, $\omega_y = \frac{V}{R}\sin\phi_0$ and $\omega_z = \frac{V}{R}\cos\phi_0$ (both positive if we assume that the aircraft is turning clockwise in the horizontal plane). By definition, the x component of the stability axis is chosen to coincide with the translational velocity in the maneuver, and therefore it is $v_x = V$. This leads to the following force equilibrium equations along the x, y, and z stability axes, respectively:

$$\begin{aligned} T\cos\theta_T - D &= 0, \\ mg\sin\phi_0 - m\frac{V^2}{R}\cos\phi_0 &= 0, \\ mg\cos\phi_0 - (L - T\sin\theta_T) + m\frac{V^2}{R}\sin\phi_0 &= 0. \end{aligned} \tag{4.62}$$

In this expression, the engine mount angle, θ_T, is given with respect to the x–y stability axis, whose definition may vary at different trim conditions. Contrary to the pull-up conditions, the equilibrium equations in a steady turn are more naturally written in vertical and horizontal axes, such that the last two equations in (4.62) become

$$\begin{aligned}(L - T\sin\theta_T)\sin\phi_0 - m\frac{V^2}{R} &= 0,\\ (L - T\sin\theta_T)\cos\phi_0 - mg &= 0.\end{aligned} \tag{4.63}$$

Steady turns are, therefore, achieved with a fixed commanded input on the vehicle controls. Euler angles are $\phi(t) = \phi_0$, $\theta(t) = 0$, and $\psi = \psi_0 + \frac{V}{R}t$, where ψ_0 is the initial heading at $t = 0$. Note that the components of the angular velocity in the body-attached stability axes satisfy Equation (4.13). From the previous expressions, it is clear that the maneuverability characteristics of an aircraft in both pull-up and turn (in particular, the achievable turning radius R) will be closely linked to the maximum available lift. The metric used here is the *load factor*, which is discussed next.

4.5.2 Load Factor and Maneuvering Envelope

We define the *load factor*, n, as the ratio between lift and weight, that is, $n = \frac{L}{mg}$ during a steady maneuver.[4] Steady maneuvers are normally defined by their load factor, and the maximum maneuvering load factor that a vehicle can sustain is a key structural sizing parameter. According to the category of the aircraft, the regulations define the load factor range, varying from values as low as 2 or 3 in the case of transport aircraft, to values as high as 8 or 9 for fighter planes. It is, however, customary to express load factors with respect to gravity, so one would talk about a "6g" maneuver. For the three conditions shown in Figure 4.10, it is easy to show that the load factors are

$$\begin{aligned}n_{\text{climb}} &= \cos\theta_0,\\ n_{\text{pull-up}} &= \cos\theta + \frac{V^2}{gR},\\ n_{\text{turn}} &= \frac{1}{\cos\phi_0}.\end{aligned} \tag{4.64}$$

Note that $n_{\text{pull-up}}$ is only defined as an instantaneous load factor, as the pitch orientation of the aircraft will vary during the maneuver. In fact, pull-up maneuvers for airframe design are defined on level flight conditions, $\theta = 0$, since this defines the maximum value of $n_{\text{pull-up}}$. Commercial transport aircraft regulations require the airframe to be able to sustain a 2.5-g pull-up from level flight. On the other extreme, a very small, or even zero, load factor can be obtained if the aircraft performs this maneuver with a negative angular velocity (*pull-down* maneuver). This is the situation for parabolic flights (zero-gravity maneuver), which generate the perception of weightlessness inside the cabin.

The maximum and minimum load factors that an aircraft can sustain, for given gross weight, configuration – e.g., clean or landing – and altitude, will depend on the airspeed. This determines the vehicle's *maneuvering envelope*, which is given as a plot of the available load factor over airspeed. This results in a V–n diagram, which

4 For transient processes (i.e., dynamic maneuvers), we use instead the instantaneous load factor, which is defined in Equation (3.110).

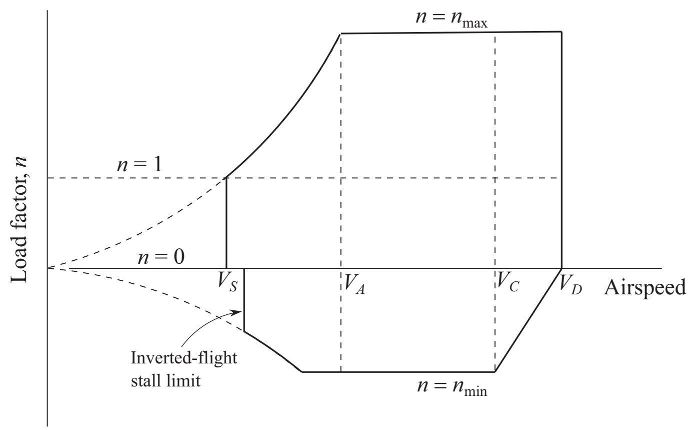

Figure 4.11 Typical maneuvering envelope.

typically looks like the one shown in Figure 4.11. Noting that a gust encounter results in a change of lift, and therefore a change of load factor, additional flight restrictions may occur when the design gust loads of Chapter 2 are also included in the V–n diagram (Niu, 1999, Ch. 3), but they are not shown in the figure. The upper and lower bounds in the load factor, $n_{\max}$ and $n_{\min}$, respectively, are structural limits; the minimum and maximum airspeeds are operational limits, that is, the stall V_S and dive V_D speed, respectively. At the lower speeds ($V < V_A$, where V_A is known as the *design maneuvering speed*, the maximum achievable load factor is determined by stall conditions ($C_{L\max}$) and similarly for the negative load factor. Note that in the horizontal axis, the speed is typically the equivalent airspeed (EAS), Equation (2.2), which makes the V-n diagram altitude independent.

The lowest speed at which the aircraft can achieve the maximum load factor is the so-called *corner speed*, V_{co}. For the clean configuration, $V_{\text{co}} = V_A$ in Figure 4.11. At this speed, one obtains the minimum turn radius, $R_{\min}$ (and the maximum attainable turn rate) at the given configuration. The minimum turn radius is derived using Equation (4.63), that is,

$$\begin{aligned} (L - T\sin\theta_T)\cos\phi_{\max} &= mg, \\ (L - T\sin\theta_T)\sin\phi_{\max} &= \frac{mV_{\text{co}}^2}{R_{\min}}, \end{aligned} \tag{4.65}$$

where the maximum roll angle is obtained from the maximum load factor in Equation (4.64), that is, $\cos\phi_{\max} = 1/n_{\max}$. As a result, we have

$$\begin{aligned} \frac{mV_{\text{co}}^2}{R_{\min}} &= \frac{mg}{\cos\phi_{\max}}\sin\phi_{\max} \\ &= mg\frac{\sqrt{1-\cos^2\phi_{\max}}}{\cos\phi_{\max}} \\ &= mg\sqrt{n_{\max}^2 - 1}, \end{aligned} \tag{4.66}$$

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and, finally,

$$R_{\min} = \frac{V_{\mathrm{co}}^2}{g\sqrt{n_{\max}^2 - 1}}. \tag{4.67}$$

Note that the results above assume that we have enough thrust and lift available to keep altitude and airspeed during the turning maneuver at the maximum load factor.

4.6 Linearized Vehicle Dynamics about a Steady Climb

Consider a trimmed rigid aircraft with leveled wings ($\phi_0 = 0$) that follows a rectilinear flight path with constant speed V and inclined at an angle θ_0 above the horizon. The aircraft is also at a certain altitude, which defines a local, and known, air density ρ. This situation is illustrated in Figure 4.12. We assume, as it is typically the case, that the aircraft is symmetric about its x–z plane, which implies that $I_{xy} = 0$ and $I_{yz} = 0$ in the inertia tensor, Equation (4.21), that is,

$$\mathbf{I}_B = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{zx} & 0 & I_{zz} \end{bmatrix}. \tag{4.68}$$

We also assume that the reference thrust, T_0, is oriented along the x axis (i.e., $\theta_T = 0$), and it does not generate pitching moments about the aircraft CM ($x_T = z_T = 0$ in Equation (4.30)), as we have already done in Section 4.5.

We are interested in investigating the small-amplitude dynamics of the aircraft about this reference trajectory. In order to first characterize the reference conditions, the trim equations of motion for this problem in the stability axes are obtained from Equations (4.59) and (4.60), yielding

$$\begin{aligned} L_0 &= mg\cos\theta_0, \\ D_0 &= T_0 - mg\sin\theta_0. \end{aligned} \tag{4.69}$$

This is the particular case of steady climb for the general equilibrium condition (4.53). The lift, drag, and thrust necessary to sustain the climb are obtained by a particular setting of the control inputs and vehicle states. Specific details of these settings, however, are not necessary for the discussion that follows. They also enforce that the resultant moment components at the CM are also zero in the reference condition.

In order to investigate the subsequent dynamic response of the aircraft to either pilot commands or atmospheric disturbance, we use the general EoM (4.26) expressed in components along the body-fixed frame (stability axes). Introducing the inertia tensor from Equation (4.68), we end up with the following six equations:

$$\begin{aligned} m(\dot{v}_x - \omega_z v_y + \omega_y v_z) &= -mg\sin\theta + f_x, \\ m(\dot{v}_y + \omega_z v_x - \omega_x v_z) &= mg\cos\theta\sin\phi + f_y, \\ m(\dot{v}_z - \omega_y v_x + \omega_x v_y) &= mg\cos\theta\cos\phi + f_z, \end{aligned} \tag{4.70}$$

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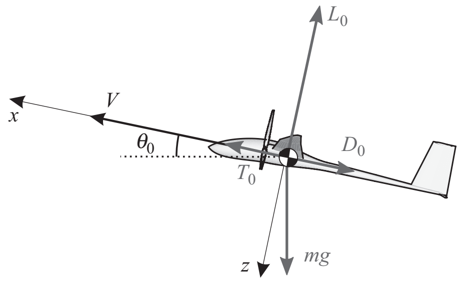

Figure 4.12 Reference conditions: steady climb.

$$
\begin{aligned}
I_{xx}\dot{\omega}_x - I_{xz}\dot{\omega}_z + (I_{zz} - I_{yy})\omega_z\omega_y - I_{xz}\omega_x\omega_y &= m_x, \\
I_{yy}\dot{\omega}_y + (I_{xx} + I_{zz})\omega_z\omega_x + I_{xz}\omega_x(\omega_x^2 - \omega_z^2) &= m_y, \\
I_{zz}\dot{\omega}_z - I_{xz}\dot{\omega}_x + (I_{yy} - I_{xx})\omega_x\omega_y + I_{xz}\omega_z\omega_y &= m_z,
\end{aligned}
$$

where the gravitational effects have been brought forward explicitly, while the aerodynamic and propulsive ones are left in terms of vector components in the stability axes in the reference condition, f_i and m_i, with $i = x, y, z$. They are solved together with the kinematic relations given by Equation (4.16), which can be rewritten as

$$
\begin{aligned}
\dot{\phi} &= \omega_x + \omega_y \sin\phi \tan\theta + \omega_z \cos\phi \tan\theta, \\
\dot{\theta} &= \omega_y \cos\phi - \omega_z \sin\phi, \\
\dot{\psi} &= \omega_y \sin\phi \sec\theta + \omega_z \cos\phi \sec\theta.
\end{aligned}
\tag{4.71}
$$

Since we are interested in the small perturbations about a reference trajectory, we can consider incremental loading from the steady-state conditions as

$$
\begin{aligned}
f_x &= T_0 - D_0 + \Delta f_x, & m_x &= \Delta m_x, \\
f_y &= \Delta f_y, & m_y &= \Delta m_y, \\
f_z &= -L_0 + \Delta f_z, & m_z &= \Delta m_z,
\end{aligned}
\tag{4.72}
$$

which results in small perturbations on the aircraft state variables from their reference values, that is,

$$
\begin{aligned}
v_x &= V + \Delta v_x, & \omega_x &= \Delta\omega_x, & \phi &= \Delta\phi, \\
v_y &= \Delta v_y, & \omega_y &= \Delta\omega_y, & \theta &= \theta_0 + \Delta\theta, \\
v_z &= \Delta v_z, & \omega_z &= \Delta\omega_z, & &
\end{aligned}
\tag{4.73}
$$

where, as before, $\Delta\bullet$ refers to small incremental changes in magnitude, for example, $\|\frac{\Delta v_x}{V}\| \ll 1$. Angles $\theta_0 + \Delta\theta$ and $\Delta\phi$ are the instantaneous climb and bank flight-path angles, respectively. Moreover, changes on the azimuth angle, $\Delta\psi$, have no effect on the aircraft dynamics and do not need to be included. The relation between sufficiently small incremental forces in Equation (4.72) and the corresponding (small) response is given by the *small-perturbation EoMs*. They are obtained by substituting Equations (4.72) and (4.73) into Equations (4.70) and (4.71), respectively, and neglecting

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nonlinear terms. The linearized EoMs become

$$
\begin{aligned}
m\Delta\dot{v}_x &= \Delta f_x - (mg\cos\theta_0)\,\Delta\theta, \\
m\Delta\dot{v}_y &= \Delta f_y + (mg\cos\theta_0)\,\Delta\phi - mV\Delta\omega_z, \\
m\Delta\dot{v}_z &= \Delta f_z - (mg\sin\theta_0)\,\Delta\theta + mV\Delta\omega_y, \\
I_{xx}\Delta\dot{\omega}_x - I_{xz}\Delta\dot{\omega}_z &= \Delta m_x, \\
I_{yy}\Delta\dot{\omega}_y &= \Delta m_y, \\
-I_{xz}\Delta\dot{\omega}_x + I_{zz}\Delta\dot{\omega}_z &= \Delta m_z,
\end{aligned}
\tag{4.74}
$$

where we have used the following trigonometric relations:

$$
\begin{aligned}
\sin\theta = \sin(\theta_0+\Delta\theta) = \sin\theta_0\cos\Delta\theta + \sin\Delta\theta\cos\theta_0 \approx \sin\theta_0 + \Delta\theta\cos\theta_0, \\
\cos\theta = \cos(\theta_0+\Delta\theta) = \cos\theta_0\cos\Delta\theta - \sin\theta_0\sin\Delta\theta \approx \cos\theta_0 - \Delta\theta\sin\theta_0.
\end{aligned}
\tag{4.75}
$$

Similarly, the linearized kinematic relations are obtained as

$$
\begin{aligned}
\Delta\dot{\phi} &= \Delta\omega_x + \Delta\omega_z\tan\theta_0, \\
\Delta\dot{\theta} &= \Delta\omega_y.
\end{aligned}
\tag{4.76}
$$

For given external forces and moments, Equations (4.74) and (4.76) define a linear system with eight equations and eight unknowns. A noticeable feature of the small-perturbation equations is that, under typical aerodynamic and propulsive features (i.e., displaying symmetry about the x–z plane), they can be split into two independent sets of equations, corresponding to the longitudinal and lateral aircraft dynamics. However, while the longitudinal dynamics are only dependent on longitudinal states, the lateral-directional dynamics are also dependent on the trim pitch and flight-path angles. They are considered in more detail below.

4.6.1 Longitudinal Problem

The longitudinal problem corresponds to aircraft kinematics in the x–z plane. It involves the translations along both axes, as well as the pitch rotation and the pitch rate, which are given by

$$
\begin{aligned}
m\Delta\dot{v}_x &= \Delta f_x - (mg\cos\theta_0)\,\Delta\theta, \\
m\Delta\dot{v}_z &= \Delta f_z - (mg\sin\theta_0)\,\Delta\theta + mV\Delta\omega_y, \\
I_{yy}\Delta\dot{\omega}_y &= \Delta m_y, \\
\Delta\dot{\theta} &= \Delta\omega_y.
\end{aligned}
\tag{4.77}
$$

The incremental forces Δf_x and Δf_z are the tangent and normal forces, respectively, to the forward-flight velocity after perturbations. To simplify this description, we assume that the thrust force has constant magnitude and its direction is always aligned with the x axis. This assumption implies that Δf_x, Δf_z, and Δm_y only depend in our model on the aerodynamic forces. The incremental aerodynamic forces resulting from perturbations of the aircraft and wind velocities have been computed in

Equations (4.36) and (4.44). For the longitudinal problem, they can be written as

$$\begin{aligned}
\Delta f_x &= \frac{1}{2}\rho V^2 S\left[C_{L0}\Delta\alpha - \Delta C_D - 2C_{D0}\frac{\Delta v_x - \Delta v_{gx}}{V}\right],\\
\Delta f_z &= \frac{1}{2}\rho V^2 S\left[-C_{D0}\Delta\alpha - \Delta C_L - 2C_{L0}\frac{\Delta v_x - \Delta v_{gx}}{V}\right],\\
\Delta m_y &= \frac{1}{2}\rho V^2 S c_{\text{ref}}\left[\Delta C_{\mathcal{M}} + 2C_{\mathcal{M}0}\frac{\Delta v_x - \Delta v_{gx}}{V}\right].
\end{aligned} \tag{4.78}$$

The incremental aerodynamic coefficients, ΔC_D, ΔC_L, and $\Delta C_{\mathcal{M}}$, can be obtained by Equations (4.41) and (4.42), with $\Delta\alpha = \frac{\Delta v_z - \Delta v_{gz}}{V}$ and $\Delta\bar{q} = \Delta\omega_y - \Delta\omega_{gy}$. Collecting the results above, we can write the incremental aerodynamic forces in the form of Equation (4.47), which becomes, for the longitudinal problem,

$$\begin{Bmatrix}\Delta f_x\\ \Delta f_z\\ \Delta m_y\end{Bmatrix} = \rho V \boldsymbol{\mathcal{A}}_{rr}\begin{Bmatrix}\Delta v_x\\ \Delta v_z\\ \Delta\omega_y\end{Bmatrix} - \rho V \boldsymbol{\mathcal{A}}_{rr}\begin{Bmatrix}\Delta v_{gx}\\ \Delta v_{gz}\\ \Delta\omega_{gy}\end{Bmatrix} + \frac{1}{2}\rho V^2 \boldsymbol{\mathcal{A}}_{ru}\delta_e, \tag{4.79}$$

with aerodynamic influence coefficient matrices defined as

$$\boldsymbol{\mathcal{A}}_{rr} = S\begin{bmatrix} -C_{D0} & \frac{1}{2}(C_{L0} - C_{D\alpha}) & -\frac{1}{2}VC_{Dq}\\ -C_{L0} & -\frac{1}{2}(C_{D0} + C_{L\alpha}) & -\frac{1}{2}VC_{Lq}\\ c_{\text{ref}}C_{\mathcal{M}0} & \frac{1}{2}c_{\text{ref}}C_{\mathcal{M}\alpha} & \frac{1}{2}Vc_{\text{ref}}C_{\mathcal{M}q}\end{bmatrix} \tag{4.80}$$

and

$$\boldsymbol{\mathcal{A}}_{ru} = S\left[-C_{D\delta_e} \quad -C_{L\delta_e} \quad c_{\text{ref}}C_{\mathcal{M}\delta_e}\right]^\top. \tag{4.81}$$

We have established the linear relation between perturbations of the aircraft flight dynamic states (linear and angular velocities in the x–z plane) and inputs (elevator deflections), on the one side, and the incremental aerodynamic forces appearing on the vehicle, on the other side. They further depend on the aircraft configuration and geometry, as well as flight conditions. All the aircraft-level longitudinal aerodynamic coefficients are defined using S and c_{ref} as normalization constants, as before. So far, we have not made any assumptions on the flow regime, and indeed Equation (4.78) is valid for low- and high-speed flight. In general, the incremental aerodynamic coefficients, ΔC_L, ΔC_D, and $\Delta C_{\mathcal{M}}$, are obtained for a given aircraft and flight regime in terms of Equations (4.41) and (4.42), with tabulated values for the stability and control derivatives. Here, following Ashley (1974), we introduce next a simplified aerodynamic model to estimate those coefficients for a conventional wing–fuselage–tail aircraft.

A Simplified Quasi-Steady Aerodynamic Model for a Conventional Configuration

For a conventional aircraft configuration, and assuming quasi-steady aerodynamics, the lift is generated both on the wing and the tail, as illustrated in Figure 4.13. Let $C^w_{L\alpha}$ and $C^t_{L\alpha}$ be the lift–curve slope of the wing and the horizontal tail, respectively, and let $C^t_{L\delta}$ be the tail-lift increment to a unit elevator input. They are typically normalized

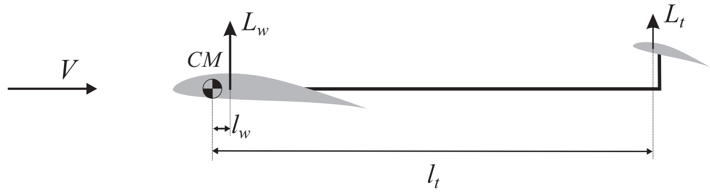

Figure 4.13 Wing–tail geometry definition.

with respect to their corresponding reference areas, that is, S_t for the tail coefficients and S for the wing coefficients. The incremental lift due to changes on the aircraft velocity becomes

$$\Delta C_L = \Delta C_L^w + \Delta C_L^t = \left(C_{L\alpha}^w + \frac{S_t}{S} C_{L\alpha}^t \right) \frac{\Delta v_z}{V} + \frac{S_t}{S} \left(C_{L\alpha}^t \frac{l_t \Delta \omega_y}{V} + C_{L\delta}^t \delta_e \right). \quad (4.82)$$

From here, we can directly identify some of the aircraft aerodynamic coefficients in Equations (4.80) and (4.81), in particular,

$$\begin{aligned} C_{L\alpha} &= C_{L\alpha}^w + \frac{S_t}{S} C_{L\alpha}^t, \\ C_{Lq} &= \frac{c_{\text{ref}}}{V} \mathbb{V}_h C_{L\alpha}^t, \\ C_{L\delta_e} &= \frac{c_{\text{ref}}}{l_t} \mathbb{V}_h C_{L\delta}^t, \end{aligned} \quad (4.83)$$

where we have introduced the nondimensional parameter $\mathbb{V}_h = \frac{S_t l_t}{S c_{\text{ref}}}$, which is known as the *horizontal tail volume coefficient*, where l_t is the distance between the aerodynamic center of the tail and the CM. The tail volume coefficient is a design parameter used to assess tail effectiveness in improving longitudinal stability (through the negative moments generated against a perturbation on the pitch rate) or controllability (through the moments generated by elevator deflections). Typically, $\mathbb{V}_h$ takes values between 0.3 and 0.6 for conventional configurations. Note that the tail aerodynamics may also include the effect of the downwash of the main wing, although this has not been explicitly written. We have finally assumed that the lift on the tail is in the same direction that the lift on the main wing, which neglects the contributions of wing downwash and pitch rate onto the instantaneous wind velocity vector on the tail. See Etkin (1972, section 6.3) for further details.

The drag coefficient is mostly independent of the forward-flight velocity at subsonic speeds. It does change, however, under changes of the lift coefficient (and, therefore, the angle of attack) through the induced drag. As a result, changes in the drag coefficient can then be shown as

$$\Delta C_D = \frac{S}{\pi b_{\text{ref}}^2 e} \left(C_L^2 - C_{L0}^2 \right) \approx \frac{S}{\pi b_{\text{ref}}^2 e} 2 C_{L0} \Delta C_L, \quad (4.84)$$

where e is the Oswald's efficiency factor ($e = 0.8$ for a rectangular wing). It has also been assumed that both e and the aspect ratio have the same value for both the wing

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and the tail, leading to

$$\Delta C_D = \frac{2SC_{L0}}{\pi b_{\text{ref}}^2 e}\left[C_{L\alpha}\frac{\Delta v_z}{V} + \frac{S_t}{S}\left(C_{L\alpha}^t\frac{l_t\Delta\omega_y}{V} + C_{L\delta}^t\delta_e\right)\right], \tag{4.85}$$

and, therefore,

$$\begin{aligned} C_{D\alpha} &= \frac{2SC_{L0}}{\pi b_{\text{ref}}^2 e}C_{L\alpha}, \\ C_{Dq} &= \frac{2SC_{L0}}{\pi b_{\text{ref}}^2 e}\frac{c_{\text{ref}}}{V}\mathbb{V}_h C_{L\alpha}^t, \\ C_{D\delta_e} &= \frac{2SC_{L0}}{\pi b_{\text{ref}}^2 e}\frac{c_{\text{ref}}}{l_t}\mathbb{V}_h C_{L\delta}^t. \end{aligned} \tag{4.86}$$

Finally, the pitch moment coefficient (positive nose up) at the aircraft CM is approximated by adding the contributions of the lift on both the wing and the tail, as

$$\begin{aligned} \Delta C_{\mathcal{M}} &= -\frac{l_w}{c_{\text{ref}}}\Delta C_L^w - \frac{l_t}{c_{\text{ref}}}\Delta C_L^t \\ &= -\left(\frac{l_w}{c_{\text{ref}}}C_{L\alpha}^w + \mathbb{V}_h C_{L\alpha}^t\right)\frac{\Delta v_z}{V} - \mathbb{V}_h C_{L\alpha}^t\frac{l_t\Delta\omega_y}{V} - \mathbb{V}_h C_{L\delta}^t\delta_e, \end{aligned} \tag{4.87}$$

where l_w is the distance between the aerodynamic center of the wing and the CM, which is indicated by the positive sign in Figure 4.13. From this, we can again identify some additional aircraft aerodynamic coefficients in Equations (4.80) and (4.81), that is,

$$\begin{aligned} C_{\mathcal{M}\alpha} &= \frac{l_w}{c_{\text{ref}}}C_{L\alpha}^w - \mathbb{V}_h C_{L\alpha}^t, \\ C_{\mathcal{M}q} &= -\frac{l_t\mathbb{V}_h}{V}C_{L\alpha}^t, \\ C_{\mathcal{M}\delta_e} &= -\mathbb{V}_h C_{L\delta}^t. \end{aligned} \tag{4.88}$$

Collecting all the results above, we write them as in Equation (4.80), which for the longitudinal problem takes the form

$$\boldsymbol{\mathcal{A}}_{rr} = S\begin{bmatrix} -C_{D0} & \left(\frac{1}{2} - \frac{SC_{L\alpha}}{\pi b_{\text{ref}}^2 e}\right)C_{L0} & -\frac{SC_{L0}}{\pi b_{\text{ref}}^2 e}\mathbb{V}_h c_{\text{ref}} C_{L\alpha}^t \\ -C_{L0} & -\frac{1}{2}\left(C_{D0} + C_{L\alpha}\right) & -\frac{1}{2}\mathbb{V}_h c_{\text{ref}} C_{L\alpha}^t \\ c_{\text{ref}}C_{\mathcal{M}0} & -\frac{1}{2}\left(l_w C_{L\alpha}^w + \mathbb{V}_h c_{\text{ref}} C_{L\alpha}^t\right) & -\frac{1}{2}\mathbb{V}_h c_{\text{ref}} l_t C_{L\alpha}^t \end{bmatrix} \tag{4.89}$$

and

$$\boldsymbol{\mathcal{A}}_{ru} = S\left[-\frac{2SC_{L0}}{\pi b_{\text{ref}}^2 e}\frac{\mathbb{V}_h c_{\text{ref}}}{l_t}C_{L\delta}^t \quad -\frac{\mathbb{V}_h c_{\text{ref}}}{l_t}C_{L\delta}^t \quad -\mathbb{V}_h c_{\text{ref}} C_{L\delta}^t\right]^\top. \tag{4.90}$$

We have established the linear relation between perturbations of the aircraft flight dynamic states (linear and angular velocities in the x–z plane) and inputs (elevator deflections), on the one side, and the incremental aerodynamic forces appearing on the vehicle, on the other side. They depend on the geometry of the aircraft

$(S, b_{\text{ref}}, l_w, S_t, l_t, e)$, stability and control derivatives, $(C^w_{L\alpha}, C^t_{L\alpha}, C^t_{L\delta})$, the reference flight conditions (ρ, V) and steady forces at that reference $(C_{L0}, C_{D0}, C_{\mathcal{M}0})$. Note that we have written the AICs in terms of both the volume coefficient $\mathbb{V}_t$ and the reference chord, c_{ref}, but they always appear as a product in Equations (4.80) and (4.81), which are uniquely defined with the geometry parameters listed above.

State Equation

We have obtained explicit expressions for the incremental aerodynamic forces, which can now be introduced in the linearized EoMs of the longitudinal problem, Equation (4.77). As we have established in Equation (4.47), the AICs related to the velocities of the aircraft are equal and opposite to those related to any wind gusts (measured at the aircraft CM). Therefore, those can also be added to the description of the aircraft dynamics, which becomes

$$\begin{aligned}
&\mathbf{M}\begin{Bmatrix}\Delta\dot{v}_x\\ \Delta\dot{v}_z\\ \Delta\dot{\omega}_y\end{Bmatrix} = (\mathbf{F}_1+\rho V\boldsymbol{\mathcal{A}}_{rr})\begin{Bmatrix}\Delta v_x\\ \Delta v_z\\ \Delta\omega_y\end{Bmatrix} + \mathbf{F}_2\Delta\theta - \rho V\boldsymbol{\mathcal{A}}_{rr}\begin{Bmatrix}\Delta v_{gx}\\ \Delta v_{gz}\\ \Delta\omega_{gy}\end{Bmatrix} + \frac{1}{2}\rho V^2\boldsymbol{\mathcal{A}}_{ru}\delta_e,\\
&\Delta\dot{\theta} = \Delta\omega_y,
\end{aligned} \tag{4.91}$$

with

$$\mathbf{M} = \begin{bmatrix} m & 0 & 0\\ 0 & m & 0\\ 0 & 0 & I_{yy}\end{bmatrix},\quad \mathbf{F}_1 = \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & mV\\ 0 & 0 & 0\end{bmatrix},\quad \text{and}\quad \mathbf{F}_2 = \begin{bmatrix}-mg\cos\theta_0\\ -mg\sin\theta_0\\ 0\end{bmatrix}. \tag{4.92}$$

For a given initial time history of elevator commands, $\delta_e(t)$, and gust disturbance velocities, $\mathbf{w}_r = -\begin{Bmatrix}\Delta v_{gx} & \Delta v_{gz} & \Delta\omega_{gy}\end{Bmatrix}^\top$, Equation (4.91) determines the (linearized) longitudinal response of the aircraft. The steady force aerodynamic coefficients, C_{L0} and C_{D0}, which have appeared in the aerodynamic influence coefficient matrices, are given from the trim state in the reference condition, Equation (4.69), and the trimmed pitching moment $C_{\mathbb{M}0} = 0$. If those are explicitly substituted into Equation (4.91), the inversion of the (diagonal) mass matrix $\mathbf{M}$ yields, after aggregating terms, the state-space form of the EoMs for the small-perturbation longitudinal aircraft flight dynamics as

$$\dot{\mathbf{x}}_r = \mathbf{A}_{rr}\mathbf{x}_r + \mathbf{B}_r\delta_e + \mathbf{E}_r\mathbf{w}_r. \tag{4.93}$$

The state vector in this problem is $\mathbf{x}_r = \begin{Bmatrix}\Delta v_x & \Delta v_z & \Delta\omega_y & \Delta\theta\end{Bmatrix}^\top$, the constant-coefficient *state matrix* is

$$\mathbf{A}_{rr} = \begin{bmatrix}
\frac{2(mg\sin\theta_0 - T_0)}{mV} & \frac{g\cos\theta_0}{V}\left(1-\frac{2SC_{L\alpha}}{\pi b^2_{\text{ref}}e}\right) & -\frac{2g\cos\theta S_t l_t C^t_{L\alpha}}{\pi b^2_{\text{ref}}eV} & -g\cos\theta_0\\
-\frac{2g\cos\theta_0}{V} & \frac{mg\sin\theta_0 - T_0}{mV} - \frac{\rho VSC_{L\alpha}}{2m} & V - \frac{\rho VS_t l_t C^t_{L\alpha}}{2m} & -g\sin\theta_0\\
0 & -\frac{\rho V\left(Sl_w C^w_{L\alpha} + S_t l_t C^t_{L\alpha}\right)}{2I_{yy}} & -\frac{\rho VS_t l_t^2 C^t_{L\alpha}}{2I_{yy}} & 0\\
0 & 0 & 1 & 0
\end{bmatrix}, \tag{4.94}$$

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the *input matrix*, which in this case is a column matrix, is

$$\mathbf{B}_r = \frac{1}{2}\rho V^2 \frac{S_t}{m} C^t_{L\delta} \begin{bmatrix} -\frac{4mg\cos\theta}{\rho V^2 \pi b^2_{\text{ref}} e} & -1 & -\frac{l_t m}{I_{yy}} & 0 \end{bmatrix}^\top, \tag{4.95}$$

and, finally, the *disturbance matrix* is given by

$$\mathbf{E}_r = \begin{bmatrix} \frac{2(mg\sin\theta_0 - T_0)}{mV} & \frac{g\cos\theta_0}{V}\left(1 - \frac{2SC_{L\alpha}}{\pi b^2_{\text{ref}} e}\right) & -\frac{2g\cos\theta S_t l_t C^t_{L\alpha}}{\pi b^2_{\text{ref}} e V} \\ -\frac{2g\cos\theta_0}{V} & \frac{mg\sin\theta_0 - T_0}{mV} - \frac{\rho V S C_{L\alpha}}{2m} & -\frac{\rho V S_t l_t C^t_{L\alpha}}{2m} \\ 0 & -\frac{\rho V\left(S l_w C^w_{L\alpha} + S_t l_t C^t_{L\alpha}\right)}{2I_{yy}} & -\frac{\rho V S_t l_t^2 C^t_{L\alpha}}{2I_{yy}} \\ 0 & 0 & 0 \end{bmatrix}. \tag{4.96}$$

We can assume finally that there are appropriate sensors on the vehicle to measure the instantaneous velocities of all states $\mathbf{x}_r$, which can support *full-state feedback*. The output equation, assuming that no feedthrough is necessary, is then the identity relation

$$\mathbf{y} = \mathbf{C}_r \mathbf{x}_r, \tag{4.97}$$

with the *output matrix* $\mathbf{C}_r = \mathcal{I}$. The eigenvalues of the state matrix determine the stability of the aircraft. In general, there will be two complex pairs, a low-frequency low-damping pair, known as the long-period (or *phugoid*) mode, and a higher frequency higher damping pair, known as the *short-period mode*, or short-period pitch oscillations (SPPOs). The values typically depend on the location of the aircraft CM (which enters the equations through l_w and l_t) and the horizontal tail area S_t. This is illustrated in Example 4.1.

Finally, we can use the methods of Section 2.3.2 to determine the statistical characteristics in the longitudinal response of an aircraft, defined by Equations (4.93) and (4.97), to atmospheric turbulence given by, for example, the von Kármán model, Equation (2.46). This is illustrated in Example 4.2. Conversely, such methods can also be used to estimate atmospheric turbulence levels from aircraft acceleration measurements during flight. This was used by Cornman et al. (1995) to propose a system in which commercial transport aircraft act as meteorological sensors by reporting load factor statistics in cruise flight, Equation (3.110), and cross-correlating them with detailed weather maps.

Example 4.1 Longitudinal Stability of a Rigid Glider. Consider a glider at an altitude of 1,000 m ($\rho = 1.112$ kg/m^3) on a descent trajectory with a glide angle $\theta_0 = -2°$ and velocity $V = 30$ m/s. The vehicle has a rectangular wing and a tail with symmetric airfoils. Furthermore, the wing is mounted at zero angle of incidence, while the tail is at $\alpha_{t0} = -5°$. Its linear dynamics is assumed to be well approximated by Equation (4.93), with the following geometrical, mass, and aerodynamic parameters,

$m = 318$ kg	$S = 6.11$ m^2
$I_{yy} = 432$ kg·m^2	$S_t = 1.14$ m^2
$C^w_{L\alpha} = 5.55$	$b_{\text{ref}} = 15$ m
$C^t_{L\alpha} = 4.30$	$l_t = 4.63$ m
$C^t_{L\delta} = 0.45$	$l_w = -0.30$ m

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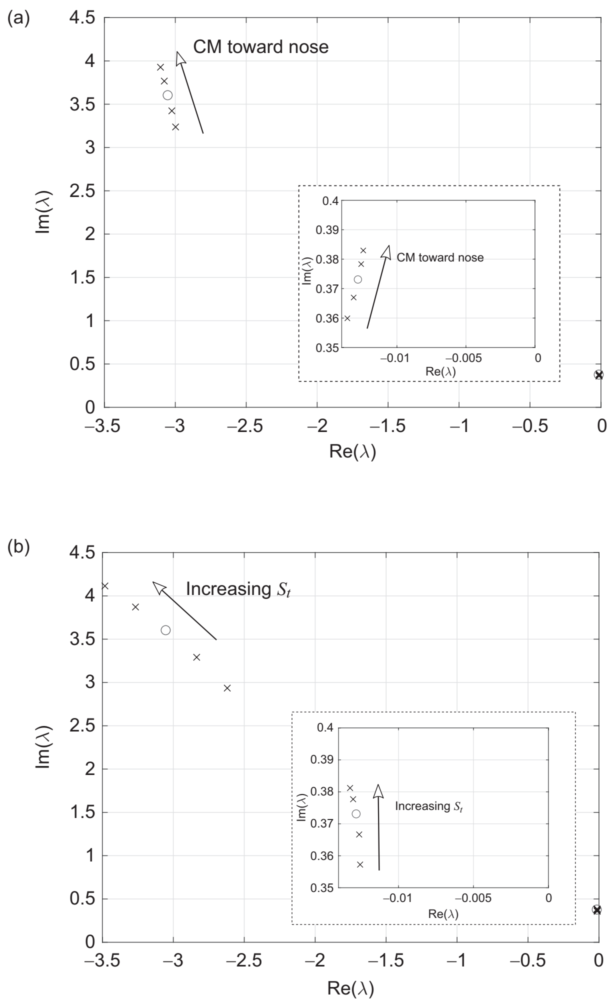

Figure 4.14 Evolution of the longitudinal eigenvalues with selected aircraft coefficients (the circle represents the baseline properties). Effects of (a) center-of-mass position and (b) the horizontal tail area, S_t. `[04RigidAircraft/glider.m]`

where, as before, the reference area S is the wing area. First, we can determine the trim condition. The load factor in trim is $n = \cos\theta_0 = 0.9994$. The value of the trim angle of attack, α_0, and the corresponding elevator deflection, δ_0, to keep the trajectory at the given path are given by the equilibrium of forces in the normal direction and the equilibrium of pitching moment, respectively, as

$$SC_{L\alpha}^{w}\alpha_0 + S_t C_{L\alpha}^{t}(\alpha_0 + \alpha_{t0}) + S_t C_{L\delta}^{t}\delta_0 = \frac{2nmg}{\rho V^2},$$
$$l_w S C_{L\alpha}^{w}\alpha_0 - l_t S_t C_{L\alpha}^{t}(\alpha_0 + \alpha_{t0}) - l_t S_t C_{L\alpha}^{t}\delta_0 = 0.$$

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Table 4.1 Eigenvectors in the reference conditions. [04RigidAircraft/glider.m]

	Short period	Phugoid
v_x/V	$0.01 \mp i0.02$	0.87
v_z/V	1	-0.11
$\omega_y l_t/V$	$-0.16 \pm i0.58$	$0.06 \mp i0.01$
θ	$0.75 \mp i0.34$	$-0.12 \mp i0.99$

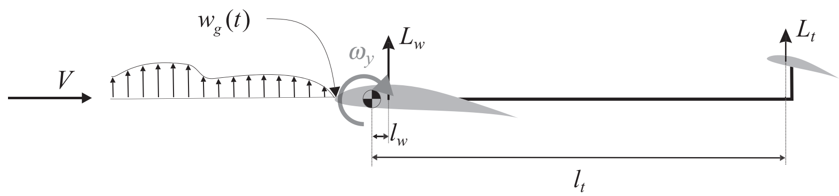

Figure 4.15 Wing–tail configuration under a vertical gust.

The numerical solution to this problem gives $\alpha_0 = 9.89°$ and $\delta_0 = -4.35°$. Next, we construct the state matrix, Equation (4.94), from which the eigenvalues (in rad/s) are obtained as $\lambda_1 = -0.013 \pm i0.373$ and $\lambda_2 = -3.05 \pm i3.60$. They correspond to a phugoid and a short-period mode, respectively, with their normalized eigenvectors, as shown in Table 4.1. They display the typical characteristics of the longitudinal modes: The phugoid is a slow exchange between kinetic (velocity) and potential energy (pitch), while the short period is a fast oscillation in pitch. The sensitivity of the eigenvalues to the location of the CM (which can be modified in the model by changing l_w while keeping $l_w + l_t$ constant) and the area of the stabilizer, S_t, are shown in Figure 4.14. The plot only shows the roots with a positive imaginary part, and their complex conjugates are also solutions. In all cases, results are for the nominal value of the parameter, marked with a circle, as well as 80%, 90%, 110%, and 120% of the reference value, which are identified with crosses. Moving the center-of-gravity aft (toward the rear of the plane, i.e., l_w growing and l_t decreasing) reduces the frequency of both flight dynamic modes, while increasing the area of the stabilizer increases both frequencies and introduces additional damping on the short-period mode. This behavior is typical of conventional wing–body–tail aircraft at subsonic speeds.

Example 4.2 Pitch Response of a Rigid Aircraft under Continuous Turbulence. Consider a rigid aircraft with a conventional configuration of Figure 4.15 and moment of inertia I_{yy} about its CM. The aircraft flies with constant velocity V through vertical (positive up) atmospheric turbulence w_g, which is approximated using the von Kármán model of Section 2.3.3. The aircraft wing, with the reference area S, has its aerodynamic center at a distance l_w ahead of the CM. Its horizontal tail, with the reference area S_t, has the resultant lift applied at a distance l_t behind the CM.

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We assume that the only relevant dynamics is the pitching of the plane,[5] with the instantaneous pitch rate $\omega_y(t)$ (positive nose up), and that the aerodynamic forces are quasi-stationary, with the lift–curve slope for the wing and tail known and given as $C^w_{L\alpha}$ and $C^t_{L\alpha}$, respectively. The EoMs for the aircraft around a certain steady-state reference condition are simply

$$I_{yy}\dot{\omega}_y = \Delta M_y = -l_w \Delta L_w - l_t \Delta L_t, \tag{4.98}$$

where ΔM_y are the instantaneous aerodynamic pitching moment at the CM during the transient dynamics, which are due to the variations of lift at both the wing and the tail. Assuming that the wing's aerodynamic center is close enough to the CM such that $l_w \left\|\omega_y\right\| \ll \left\|w_g\right\|$, the lift on the wing is simply due to the change of the effective angle of attack as the atmospheric turbulence impacts the wing, that is,

$$\Delta L_w = \tfrac{1}{2}\rho V^2 S C^w_{L\alpha} \frac{w_g(t)}{V}. \tag{4.99}$$

The aerodynamic forces on the tail have contributions from the atmospheric turbulence and from the instantaneous pitch rate, that is,

$$\Delta L_t = \tfrac{1}{2}\rho V^2 S_t C^t_{L\alpha} \frac{w_g(t) + l_t\omega_y(t)}{V}. \tag{4.100}$$

Here we have neglected the time that the fluid needs to move between the wing and the tail, $\frac{l_t - l_w}{V}$, which can result in a small delay if either V or the turbulent length scale is sufficiently small (as, e.g., in low-altitude flight). Substituting Equations (4.99) and (4.100) in the EoM yields,

$$I_{yy}\dot{\omega}_y + \tfrac{1}{2}\rho V^2 S_t l_t C^t_{L\alpha} \frac{l_t\omega_y}{V} = -\tfrac{1}{2}\rho V^2 \left(S l_w C^w_{L\alpha} + S_t l_t C^t_{L\alpha}\right) \frac{w_g(t)}{V}. \tag{4.101}$$

This equation can be nondimensionalized using $\hat{\omega}_y = \frac{l_t\omega_y}{V}$ and $\hat{w}_g = \frac{w_g}{V}$. The time can also be normalized by the characteristic period $T = a\ell/V$ in the atmospheric turbulence introduced in Equation (2.48). This results in

$$T\frac{\mathrm{d}\hat{\omega}_y}{\mathrm{d}t} + k_t\hat{\omega}_y = -\left(k_w + k_t\right)\hat{w}_g, \tag{4.102}$$

where

$$k_t = \mu_{yy}^{-1}\frac{S_t}{S}\frac{a\ell}{l_t}C^t_{L\alpha} \quad \text{and} \quad k_w = \mu_{yy}^{-1}\frac{l_w}{l_t}\frac{a\ell}{l_t}C^w_{L\alpha}, \tag{4.103}$$

with the pitch inertia parameter $\mu_{yy} = \frac{2I_{yy}}{\rho S l_t^3}$. The corresponding frequency-response function is then

$$G_{\hat{\omega}_y\hat{w}_g}(\mathrm{i}\omega) = -\frac{k_w + k_t}{k_t + \mathrm{i}\omega T}, \tag{4.104}$$

which can be combined with the FRF of the von Kármán turbulence model, Equation (2.48), to determine the statistics of the vehicle response. This is the situation

[5] The solution to the general longitudinal problem can be found in Buck and Newman (2006).

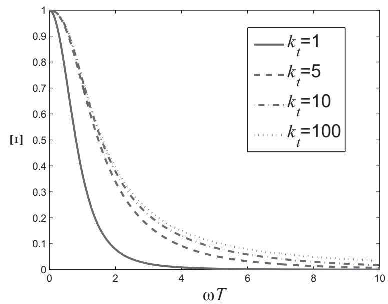

Figure 4.16 Normalized PSD of the pitch response to von Kármán turbulence, Equation (4.108). [04RigidAircraft/pitchpsd.m]

depicted in Figure 2.13, which results here in a combined FRF given by

$$G_{\omega_y n} = G_{\omega_y w_g} G_{w_g n} = \frac{1}{l_t} G_{\hat{\omega}_y \hat{w}_g} G_{w_g n} = -\frac{\sigma_w}{l_t}\sqrt{\frac{T}{\pi a}}\,\frac{k_w + k_t}{k_t + \mathrm{i}\omega T}\,\frac{1 + \frac{2\sqrt{2}}{\sqrt{3}}\mathrm{i}\omega T}{(1+\mathrm{i}\omega T)^{11/6}}. \tag{4.105}$$

The PSD of the pitch response is defined as $\Phi_{\omega_y}(\omega) = \left|G_{\omega_y n}(\mathrm{i}\omega)\right|^2 \Phi_n$, where Φ_n has, as defined above, a unit amplitude. As a result, we finally have

$$\Phi_{\omega_y}(\omega) = \frac{\sigma_w^2 T}{\pi a l_t^2}\,\frac{(k_w + k_t)^2}{k_t^2 + \omega^2 T^2}\,\frac{1 + \frac{8}{3}\omega^2 T^2}{(1+\omega^2 T^2)^{11/6}}, \tag{4.106}$$

which can also be written as

$$\Phi_{\omega_y}(\omega) = \frac{\sigma_w^2 T}{\pi a l_t^2}\left[\frac{k_w}{k_t} + 1\right]^2 \Xi(\omega T), \tag{4.107}$$

with

$$\Xi(\omega T) = \frac{k_t^2}{k_t^2 + \omega^2 T^2}\,\frac{1 + \frac{8}{3}\omega^2 T^2}{(1+\omega^2 T^2)^{11/6}}. \tag{4.108}$$

As a result, the intensity of the pitch disturbance grows with the intensity of the gust, σ_w, and with the ratio $\frac{k_w}{k_t} = \frac{S l_w C_{L\alpha}^w}{S_t l_t C_{L\alpha}^t}$, which depends only on the aircraft geometry. A large distance between the tail and the CM, l_t, results in a smaller amplitude of the pitch dynamics. The nondimensional frequency content is given by the term $\Xi(\omega T)$ in the equation and is shown in Figure 4.16 as a function of the constant k_t. Increasing the value of k_t, which can be achieved by either a lower aircraft pitch inertia or a smaller wing area, increases the bandwidth of the response.

4.6.2 Lateral Problem

Similar to the longitudinal problem, the lateral (also known as the *lateral-directional*) problem is defined by the equations in lateral translation, roll and yaw rotations, and their rates. From Equation (4.74), we obtain

$$
\begin{aligned}
m\Delta\dot{v}_y &= \Delta f_y + (mg\cos\theta_0)\,\Delta\phi - mV\Delta\omega_z,\\
I_{xx}\Delta\dot{\omega}_x - I_{xz}\Delta\dot{\omega}_z &= \Delta m_x,\\
-I_{xz}\Delta\dot{\omega}_x + I_{zz}\Delta\dot{\omega}_z &= \Delta m_z.
\end{aligned}
\tag{4.109}
$$

The incremental force Δf_y corresponds to the lateral force normal to the forward-flight velocity after perturbation to the lateral translation and roll angle,[6] while the incremental Δm_x and Δm_z moments correspond to the roll and yaw moments, respectively, after the perturbations. Other than θ_0 be nonzero (reference condition associated with the longitudinal trim), the other relevant reference conditions (i.e., $C_{Y0}, C_{\mathcal{N}0}$, and $C_{\mathcal{L}0}$) are zero (assuming symmetric aircraft) for the steady straight path case being considered here. Again, we aim to compute how the aerodynamic force and moments change with the perturbations of the states from the reference condition. The perturbations of the states of interest are a subset of Equation (4.73) and given by

$$
\begin{aligned}
v_y &= \Delta v_y,\\
\omega_x &= \Delta\omega_x, \qquad \phi = \Delta\phi,\\
\omega_z &= \Delta\omega_z.
\end{aligned}
\tag{4.110}
$$

where the perturbation in lateral velocity leads to the sideslip angle $\Delta\beta = \frac{\Delta v_y}{V}$.

To evaluate the incremental lateral force, we refer again to the general case given in Equation (4.44), and from it, we obtain

$$
\begin{aligned}
\Delta f_y &= \frac{1}{2}\rho V^2 S \Delta C_Y\\
&= \frac{1}{2}\rho V^2 S\left(C_{Y\beta}\frac{\Delta v_y}{V} + C_{Yp}\Delta\omega_x + C_{Yr}\Delta\omega_z + C_{Y\delta_a}\Delta\delta_a + C_{Y\delta_r}\delta_r\right),
\end{aligned}
\tag{4.111}
$$

where we used the lateral force coefficient as defined in Equation (4.42). Similarly, the incremental moments can be written as

$$
\begin{aligned}
\Delta m_x =& \frac{1}{2}\rho V^2 S b_{\mathrm{ref}} \Delta C_{\mathcal{N}}\\
=& \frac{1}{2}\rho V^2 S b_{\mathrm{ref}}\left(C_{\mathcal{N}\beta}\frac{\Delta v_y}{V} + C_{\mathcal{N}p}\Delta\omega_x + C_{\mathcal{N}r}\Delta\omega_z + C_{\mathcal{N}\delta_a}\delta_a + C_{\mathcal{N}\delta_r}\delta_r\right),\\
\Delta m_z =& \frac{1}{2}\rho V^2 S b_{\mathrm{ref}} \Delta C_{\mathcal{L}}\\
=& \frac{1}{2}\rho V^2 S b_{\mathrm{ref}}\left(C_{\mathcal{L}\beta}\frac{\Delta v_y}{V} + C_{\mathcal{L}p}\Delta\omega_x + C_{\mathcal{L}r}\Delta\omega_z + C_{\mathcal{L}\delta_a}\delta_a + C_{\mathcal{L}\delta_r}\delta_r\right).
\end{aligned}
\tag{4.112}
$$

From these results, we can write the incremental aerodynamic forces in the form of Equation (4.47) that becomes, for the lateral problem,

$$
\begin{Bmatrix} \Delta f_y \\ \Delta m_x \\ \Delta m_z \end{Bmatrix} = \rho V \boldsymbol{\mathcal{A}}_{rr} \begin{Bmatrix} \Delta v_y \\ \Delta\omega_x \\ \Delta\omega_z \end{Bmatrix} + \frac{1}{2}\rho V^2 \boldsymbol{\mathcal{A}}_{ru} \begin{Bmatrix} \delta_a \\ \delta_r \end{Bmatrix},
\tag{4.113}
$$

[6] The azimuth angle ψ is immaterial to the dynamic characterization of the vehicle as discussed in Section 4.6 and it does not need to be included here.

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with aerodynamic influence coefficient matrices defined as

$$\boldsymbol{\mathcal{A}}_{rr} = S \begin{bmatrix} \frac{1}{2} C_{Y\beta} & \frac{V}{2} C_{Yp} & \frac{V}{2} C_{Yr} \\ \frac{b_{\text{ref}}}{2} C_{\mathcal{N}\beta} & \frac{Vb_{\text{ref}}}{2} C_{\mathcal{N}p} & \frac{Vb_{\text{ref}}}{2} C_{\mathcal{N}r} \\ \frac{b_{\text{ref}}}{2} C_{\mathcal{L}\beta} & \frac{Vb_{\text{ref}}}{2} C_{\mathcal{L}p} & \frac{Vb_{\text{ref}}}{2} C_{\mathcal{L}r} \end{bmatrix} \tag{4.114}$$

and

$$\boldsymbol{\mathcal{A}}_{ru} = S \begin{bmatrix} C_{Y\delta_a} & C_{Y\delta_r} \\ b_{\text{ref}} C_{\mathcal{N}\delta_a} & b_{\text{ref}} C_{\mathcal{N}\delta_r} \\ b_{\text{ref}} C_{\mathcal{L}\delta_a} & b_{\text{ref}} C_{\mathcal{L}\delta_r} \end{bmatrix}. \tag{4.115}$$

Corresponding state equations can be obtained here, similar to what was done for the longitudinal problem, with respect to the state vector $\mathbf{x}_r = \{\Delta v_y \;\; \Delta\omega_x \;\; \Delta\omega_z \;\; \Delta\phi\}^\top$.

4.7 Summary

This chapter has reviewed the basic theories and fundamental engineering concepts that describe the flight dynamics of rigid aircraft. This forms the basis for the more general formulations for flexible aircraft that will be introduced in the following chapters. Importantly, the traditional descriptions in flight dynamics have been slightly modified such that it can be easily expanded to include the additional DoFs on a flexible aircraft. The focus has also been on state-space formulations, and we have heavily used notations of matrix algebra whenever possible – this not only will easy our way in future chapters but also is a good practice to integrate traditional flight dynamics analysis into techniques of linear (and nonlinear) systems theory. Finally, we have put special emphasis, as we do throughout the book, on flight under nonstationary atmospheric conditions.

The chapter has started with the definitions of the reference frames that are needed to describe the kinematics of a rigid aircraft, and Euler angles were used to parameterize its rotations. Other parameterizations are possible; some of them will be later introduced in Chapter 9 to describe the large deformations on flexible vehicles. The EoMs have then been written in the stability axes, as it is customary in flight dynamics. The main features of gravitational, propulsive, and aerodynamic forces acting on the aircraft have also been outlined.

Once the general framework to describe and analyze the aircraft dynamics has been assembled, we have looked into two relevant situations. First, the constant-velocity solutions of the differential equations have defined steady (or equilibrium) maneuvers. Those are the basic building blocks to define the flight performance of a vehicle and have generalized the definition of the load factor that we had already introduced in Section 3.6. Finally, we have looked into the small-perturbation dynamics of a rigid aircraft about a steady straight path. This problem will reappear later in this book with additional physics. A glider example has been to investigate longitudinal response to atmospheric turbulence, using the tools of Section 2.3.

4.8 Problems

Exercise 4.1. The instantaneous angular velocity can be written in terms of its components in the inertial reference frame as $\boldsymbol{\omega}_E = \mathbf{R}_{BE}\boldsymbol{\omega}_B$.

(i) Show that $\tilde{\boldsymbol{\omega}}_E = \dot{\mathbf{R}}_{EB}\mathbf{R}_{BE}$.
(ii) Using the previous result, show then that $\tilde{\boldsymbol{\omega}}_E = \mathbf{R}_{EB}\tilde{\boldsymbol{\omega}}_B\mathbf{R}_{BE}$.

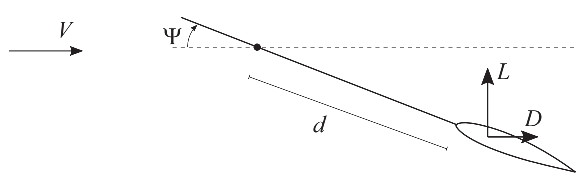
Exercise 4.2. Derive the expression of Newton–Euler equations in body axes, Equation (4.26), from the conservation properties in the Earth axes, Equation (4.24). How do they change if the body-attached frame of reference is not at the CM?

Exercise 4.3. An aircraft carrier is heading north at a certain constant speed V_s. We define a frame of reference linked to the ship, with x_s pointing north and z_s vertical up. An aircraft is also flying north with a constant forward-flight velocity, V_∞. From radar measurements, it is known that at $t = 0$, the aircraft CM is at $[x_s(0) = x_{s0}, y_s(0) = y_{s0}, z_s(0) = z_{s0}]$ with respect to the ship. The subsequent history of CM velocities is then recorded from on-board equipment (body-fixed axes), and it is known to be $\mathbf{V}_B(t)$ and $\boldsymbol{\Omega}_B(t)$.

(i) Write the equations that determine the position of the aircraft in the ship frame at a certain time $t = T$, with $T > 0$.
(ii) Determine a_y and a_z and the landing trajectory if $\boldsymbol{\Omega}_B = 0$ and the components of the translational velocity, in the B frame, follow the profile

$$
\begin{aligned}
V_x &= V_s + V_\infty \cos\frac{\pi t}{2T},\\
V_y &= a_y \cos\frac{\pi t}{2T},\\
V_z &= a_z \cos\frac{\pi t}{2T}.
\end{aligned}
$$

Exercise 4.4. After Ashley (1974), the weathervane of the figure rotates about an axis perpendicular to the plane with a large angle, ψ. It has a moment of inertia I_{zz} about that axis. Assume that the drag is negligible and that the lift is always perpendicular to the constant wind velocity and proportional to $\sin\Psi$. Rotations around the hinge of the weathervane have a viscous damping moment proportional to $\dot{\Psi}$.



(i) Show that the EoMs have the following structure:

$$\ddot{\Psi} + a_1\dot{\Psi} + a_2 \sin 2\Psi = 0.$$

(ii) Write the problem in a nonlinear state-space form as a function on a_1 and a_2.

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(iii) Using a forward-Euler discretization, that is, $\dot{\mathbf{x}}(t_n) = \frac{\mathbf{x}(t_{n+1})-\mathbf{x}(t_n)}{t_{n+1}-t_n}$, write a numerical algorithm to solve the state-space equations for the given initial conditions $\Psi(0) = \Psi_0$ and $\dot{\Psi}(0) = 0$.

(iv) Choose values for the parameters and explore the solutions to this problem numerically.

Exercise 4.5. Consider an aircraft with the maneuver envelope of Figure 4.11.

(i) For leveled turns, plot its maximum turn rate $\dot{\Psi}_{\max}$ against the flight speed.

(ii) A rectangular wing with chord c and semispan b is modeled using lifting line theory and airfoils with the known $c_{l\alpha}$ and $c_{l\delta}$. If the maximum trailing-edge flap deflection is $\delta_{\max}$, determine the size and location of the best flaps to achieve a 2% increment in lift.

Exercise 4.6. Introduce the nondimensional time $\bar{t} = \frac{gt}{V_\infty}$ and the nondimensional state variables $\bar{v}_x = \frac{v_x}{V_\infty}$, $\bar{v}_z = \frac{v_z}{V_\infty}$, and $\bar{\omega}_y = \frac{\omega_y l_t}{V_\infty}$ in the longitudinal EoMs for a rigid aircraft and show that the corresponding state matrix in a nondimensional form can be written as

$$\bar{\mathbf{A}}_r = \begin{bmatrix} 2\left(\sin\Theta_0 - \bar{T}\right) & \cos\Theta_0\left(1 - \frac{2SC_{L\alpha}}{\pi b^2 e}\right) & 0 & -\cos\Theta_0 \\ -2\cos\Theta_0 & \sin\Theta_0 - \bar{q}C_{L\alpha} - \bar{T} & \frac{V_\infty^2}{gl_t} - \bar{q}\bar{S}_t C_{L\alpha}^t & -\sin\Theta_0 \\ 0 & -\bar{q}\bar{l}_t\left[\bar{l}_w C_{L\alpha}^w + \bar{S}_t\bar{l}_t C_{L\alpha}^t\right] & -\bar{q}\bar{l}_t^2\bar{S}_t C_{L\alpha}^t & 0 \\ 0 & 0 & \frac{V_\infty^2}{gl_t} & 0 \end{bmatrix},$$

with $\bar{T} = \frac{T_0}{mg}$, $\bar{q} = \frac{q_\infty S}{mg}$, $\bar{S}_t = \frac{S_t}{S}$, $\bar{l}_X = \frac{l_X}{r_y}$, and $r_y = \sqrt{\frac{I_{yy}}{m}}$.

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